Math Bootcamp

Day 3 — Remaining real analysis and linear algebra

Math Camp 2026

How to use these slides

Main deck: definitions, recipes, examples, and practice.

Appendix and proofs (marked on relevant slides): **for interest only** — optional recap for those already comfortable with the math.

Not required for Math Camp. Focus on the main slides and exercises unless you want the extra detail.

3.1 Fixed points — equilibrium as a solution

Fixed point. A point x^* is a **fixed point** of a map T if $T(x^*) = x^*$.

Economics reading. Many equilibria are fixed-point problems:

- **Market clearing:** a price p^* with excess demand $E(p^*) = 0$.
- **Game theory:** a strategy profile that is a best response to itself (Nash).

Two useful existence tools (today).

- **Contraction Mapping Theorem:** existence and uniqueness of a fixed point; iteration converges.
- **Brouwer:** continuous map on a closed interval / convex set.

3.1 Contraction Mapping Theorem

Contraction. A map T **shrinks distances** if there is $k \in (0, 1)$ such that $|T(x) - T(y)| \leq k|x - y|$ for all x, y in the domain.

Contraction Mapping Theorem. If $T\colon [a, b] \to [a, b]$ is a contraction, then

- **Existence:** there is a fixed point x^* with $T(x^*) = x^*$.
- **Uniqueness:** that fixed point is **unique**.
- **Iteration:** for any starting value $x_0 \in [a, b]$, the sequence $x_n = T(x_{n-1})$ converges to x^*.

3.1 Why iteration converges

Step 1 — Start anywhere. Pick any x_0. Define $x_n = T(x_{n-1})$ for $n = 1, 2, \ldots$

Step 2 — Jumps shrink exponentially. Each new step is at most k times the previous jump:

$$|x_{n+1} - x_n| = |T(x_n) - T(x_{n-1})| \le k|x_n - x_{n-1}| \le k^n|x_1 - x_0|.$$

Since $k < 1$, the moves get **exponentially smaller** — the path barely changes after a while.

Step 3 — The sequence converges. Step 2 implies $x_n \to x^*$ for some limit x^* (check it is a **Cauchy sequence** for a rigorous proof). So iteration from **any** starting point drives x_n toward the same x^*.

Step 4 — The limit is a fixed point. T is continuous, so taking limits in $x_n = T(x_{n-1})$ gives $x^* = \lim x_n = \lim T(x_{n-1}) = T(\lim x_{n-1}) = T(x^*)$.

3.1 Walrasian price adjustment

Question. Excess demand $E(p)$, market clearing $E(p^*) = 0$. Prices update by

$$p_{t+1} = T(p_t) = p_t + \lambda\big(1 - e^{-\alpha E(p_t)}\big), \qquad \lambda = \tfrac{1}{5},\ \alpha = 0.1.$$

Does the price stabilize? That is, does $p_t \to p^*$?

Fixed points of T are exactly the market-clearing prices: $E(p^*) = 0 \iff T(p^*) = p^*$. Near $E = 0$, $1 - e^{-\alpha E} \approx \alpha E$.

Answer. On a compact range that T maps into itself, if $E' < 0$ and $T'(p) > -1$ (equivalently $\lambda\alpha\, e^{-\alpha E(p)}\,|E'(p)| < 2$), then yes: T is a contraction, so $p_t \to p^*$ from any start in the range.

3.1 Why the price stabilizes

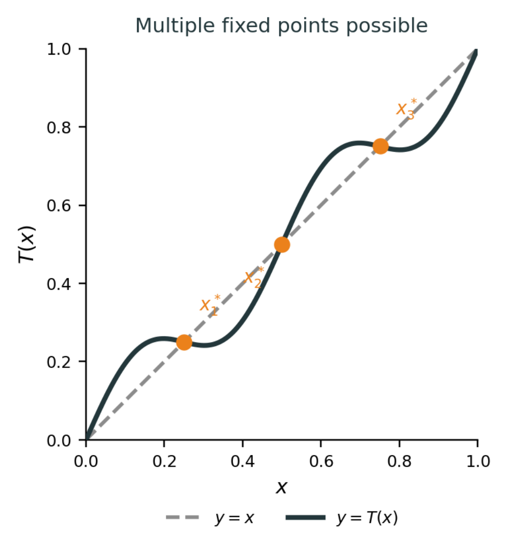
Slope. $T'(p) = 1 + \lambda\alpha\, e^{-\alpha E(p)}\, E'(p)$.

Why $T' < 1$. $E' < 0$, so the second term is negative and $T' < 1$.

Why $T' > -1$ is enough. Then $-1 < T' < 1$, so $\max|T'| = k < 1$. By MVT, $|T(p) - T(q)| = |T'(c)|\,|p - q| \leq k|p - q|$: each update shrinks the gap to p^*, hence the price stabilizes.

Brouwer (one dimension). If $T\colon [a,b] \to [a,b]$ is **continuous**, then T has **at least one** fixed point x^* with $T(x^*) = x^*$.

Picture. Plot $y = T(x)$ against the 45-degree line $y = x$. A crossing is a fixed point.



Multiple fixed points possible
1.0
0.8
0.6
0.4
0.2
0.0
T(x)
0.0
0.2
0.4
0.6
0.8
1.0
x
x_1^*
x_2^*
x_3^*
$y = x$
$y = T(x)$

3.1 Brouwer — general statement

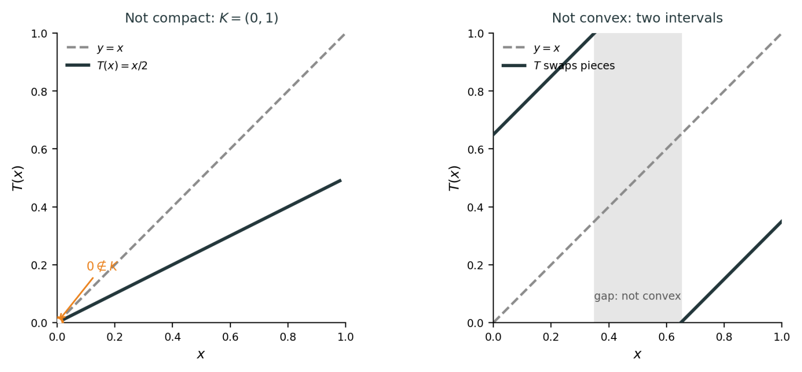
Brouwer (several variables). Let $K \subset \mathbb{R}^n$ be **nonempty**, **compact**, and **convex**. If $T : K \to K$ is **continuous**, then T has at least one fixed point $x^* \in K$ with $T(x^*) = x^*$.

Camp vocabulary.

- **Compact** in $\mathbb{R}^n$: closed and bounded (Heine–Borel).
- **Convex**: the segment between any two points in K stays in K.

One dimension. $K = [a, b]$ is compact and convex, so the 1D case is a special case.

3.1 Why compact and convex



Left: $K = (0, 1)$ not closed, $T(x) = x/2$; the crossing is at $0 \notin K$. Right: K is two intervals, T swaps them; no crossing on K.

3.1 Linear Cournot — best response

Example. Two firms, $p = a - (q_1 + q_2)$, cost 0, $q_i \in [0, a]$, $a > 0$. Firm 1 takes q_2 as given: $\pi_1(q_1; q_2) = (a - q_1 - q_2)q_1$. Firm 2 is symmetric.

FOC. $\partial\pi_1/\partial q_1 = a - 2q_1 - q_2 = 0$, so $q_1 = (a - q_2)/2$. **SOC.** $\partial^2\pi_1/\partial q_1^2 = -2 < 0$.

Best response. $BR_1(q_2) = (a - q_2)/2$, $BR_2(q_1) = (a - q_1)/2$, and both map $[0, a]$ into itself.

3.1 Linear Cournot — Nash

Nash. $q_1^* = BR_1(q_2^*)$ and $q_2^* = BR_2(q_1^*)$:

$$2q_1^* = a - q_2^*, \qquad 2q_2^* = a - q_1^*.$$

Substitute $q_2^* = a - 2q_1^*$ into the second equation: $2(a - 2q_1^*) = a - q_1^*$, so $2a - 4q_1^* = a - q_1^*$, hence $a = 3q_1^*$.

Thus $q_1^* = q_2^* = a/3$. Unique solution of two linear equations — no Brouwer needed.

Same map: $T(q_1, q_2) = \big(BR_1(q_2), BR_2(q_1)\big)$ on $K = [0, a]^2$, with unique fixed point $(a/3, a/3)$.

3.1 Brouwer — Cournot duopoly

Asymmetric nonlinear demand. Two firms, $q_i \in [0, \bar{q}]$, $p = a - (q_1 + 2q_2)^2$ (e.g. $a = 8$, $\bar{q} = 2$). Profit $\pi_i = p\, q_i$.

Interior FOCs.

$$a - 3q_1^2 - 8q_1 q_2 - 4q_2^2 = 0, \qquad a - q_1^2 - 8q_1 q_2 - 12q_2^2 = 0.$$

The two equations are not symmetric — no formula like $q_i^* = a/3$.

Existence. $T(q_1, q_2) = \big(BR_1(q_2), BR_2(q_1)\big)$ is continuous on the compact convex set $K = [0, \bar{q}]^2$. Brouwer: a Nash equilibrium exists.

3.2 Separating Hyperplane Theorem

Setup. Nonempty **convex** sets $A, B \subset \mathbb{R}^n$ with $A \cap B = \varnothing$.

Theorem. If at least one of A, B is **compact**, then there exist $\mathbf{p} \neq \mathbf{0}$ and $\beta \in \mathbb{R}$ such that

$$\mathbf{p} \cdot a \leq \beta \leq \mathbf{p} \cdot b \qquad \text{for all } a \in A,\ \ b \in B.$$

Geometrically, the hyperplane $\mathbf{p} \cdot x = \beta$ separates A from B. (If only B is compact, swap the labels of A and B.)

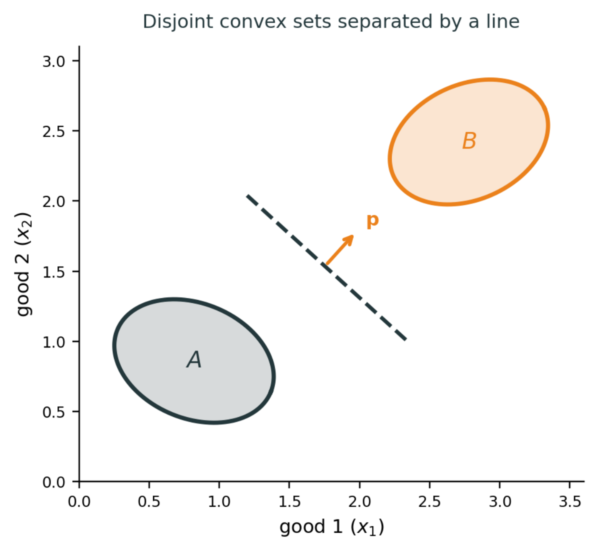
Proof and strict separation: appendix (**for interest only**).

3.2 Separating Hyperplane Theorem — two goods

With $x = (x_1, x_2)$, separation is a line $p_1 x_1 + p_2 x_2 = \beta$: all of A on one side, all of B on the other.

Two useful special cases.

- **Single point.** $B = \{b_0\}$ is compact. If $b_0 \notin A$ and A is closed and convex, a separator exists even when A is not compact.
- **Strict separation.** If the sets are a positive distance apart, the same $\mathbf{p}$ can separate **strictly**: $\mathbf{p} \cdot a < \beta < \mathbf{p} \cdot b$.


Disjoint convex sets separated by a line
good 2 (x_2)
good 1 (x_1)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
A
B
p

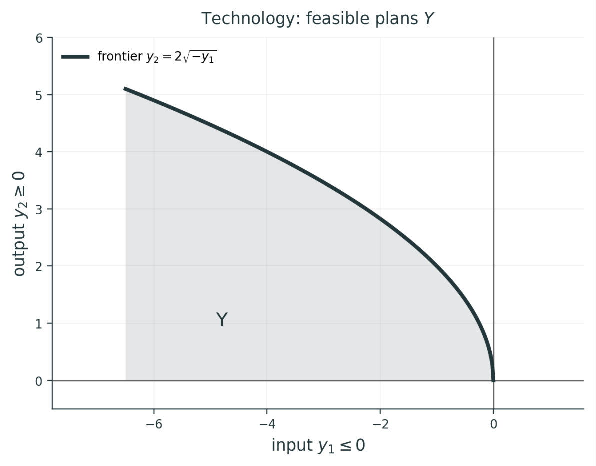
3.2 Technology

Netput. $y = (y_1, y_2)$ with input $y_1 \le 0$ and output $y_2 \ge 0$.

Technology. Labor $L = -y_1$, max output $f(L) = 2\sqrt{L}$:

$$Y = \{y : \ y_1 \le 0,\ 0 \le y_2 \le 2\sqrt{-y_1}\}.$$

Shaded: feasible plans. The frontier is **technically efficient**: you cannot get more output from the same input.

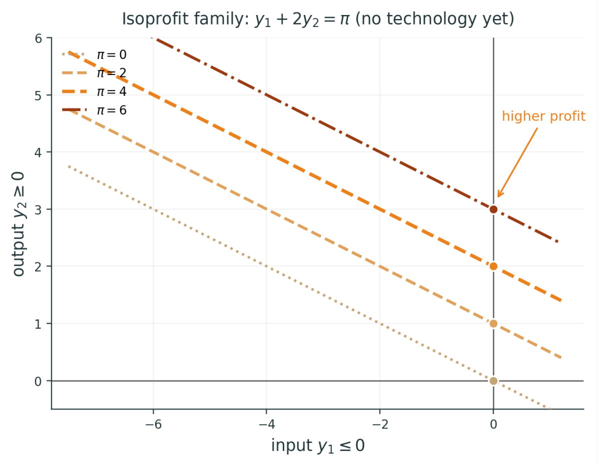


3.2 Isoprofit

Prices. Wage $p_1 = 1$, output price $p_2 = 2$. Profit $\pi = p_1 y_1 + p_2 y_2 = y_1 + 2y_2$.

Isoprofit lines. Same $\pi \Rightarrow$ a line $y_1 + 2y_2 = \pi$. Slope $-p_1/p_2 = -\frac{1}{2}$; intercept π/p_2.

Parallel lines. A higher intercept is a higher profit — this is only about π, not about Y.

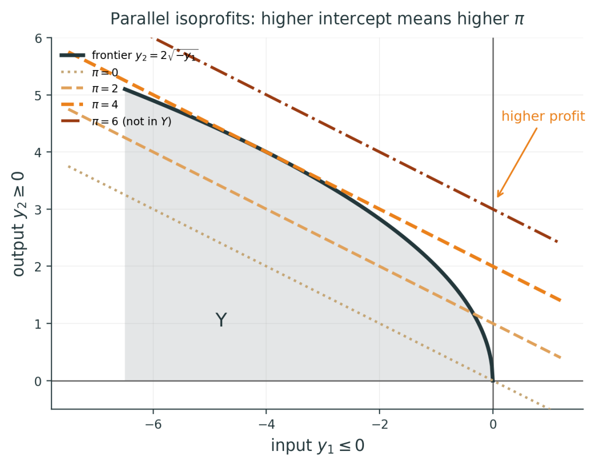


3.2 Which profits are feasible

Now put the isoprofits on the technology Y.

$\pi = 0$ and $\pi = 2$ cut through Y (attainable). $\pi = 6$ misses Y (not feasible).

The firm wants the highest π that still meets Y.

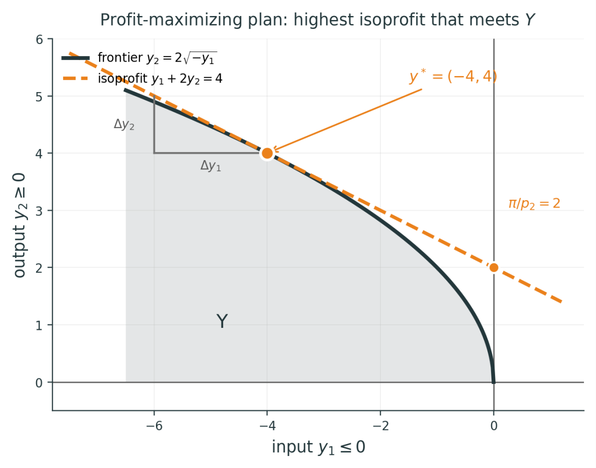


3.2 Profit-maximizing plan

Profit max. The highest isoprofit that still meets Y is tangent at $y^* = (-4, 4)$, $\pi = 4$.

That is the **profit-maximizing** plan at these prices.

y^* also lies on the frontier, so it is **technically efficient** as well. Two properties, one point.



3.2 Why the isoprofit is tangent

Suppose locally you can produce Δy_2 using Δy_1.

Extra profit is $p_1 \Delta y_1 + p_2 \Delta y_2$. At a maximizer this must be zero:

$$p_1 \Delta y_1 + p_2 \Delta y_2 = 0.$$

If it were positive you would do a bit more of that swap; if negative, a bit less. So the point would not be profit-maximizing.

Rearrange:

$$\frac{\Delta y_2}{\Delta y_1} = -\frac{p_1}{p_2}.$$

The tangent of the production frontier equals the price ratio.

3.2 Example — producer theory

Setup. A firm's **production possibility set** $Y \subset \mathbb{R}^n$ is **convex** (and closed in full treatments).

Statement. If Y is convex, the theorem guarantees that every **technically efficient** production plan $y \in Y$ is also a **profit-maximizing** choice for some nonzero price vector $\mathbf{p}$:

$$\mathbf{p} \cdot y \geq \mathbf{p} \cdot y' \quad \text{for all } y' \in Y.$$

Separation picture. Let B be outputs **strictly better than** y (more of some good, no less of any). If y is efficient, $Y \cap B = \varnothing$. A separating hyperplane gives supporting prices $\mathbf{p}$ at y.

Proof: appendix (**for interest only**).

Linear Algebra Basics

3.3 Vectors in $\mathbb{R}^n$

Vector. An ordered list of n real numbers. We write a **column vector** as

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n.$$

Its **transpose** $\mathbf{x}^\mathsf{T} = (x_1, \ldots, x_n)$ is a **row vector**.

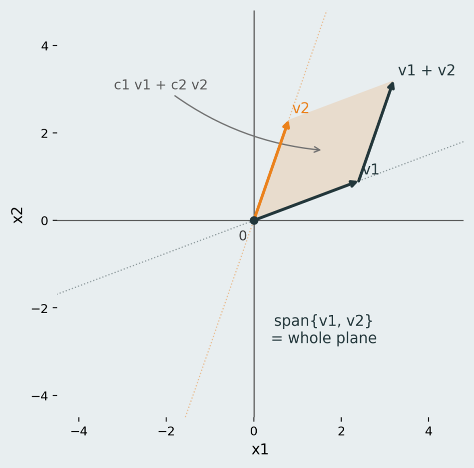
Linear combination. Given vectors $\mathbf{v}_1, \ldots, \mathbf{v}_k \in \mathbb{R}^n$ and scalars $c_1, \ldots, c_k$,

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k$$

is a **linear combination** of the $\mathbf{v}_j$.

Span. All linear combinations of $\mathbf{v}_1, \ldots, \mathbf{v}_k$ form the **span** of those vectors.

3.3 Span in the plane (two vectors)



3.3 Matrix–vector product — columns of A

Setup. $A \in \mathbb{R}^{m\times n}$ has columns $\mathbf{a}_1, \ldots, \mathbf{a}_n \in \mathbb{R}^m$ and $\mathbf{x} \in \mathbb{R}^n$ is a column vector.

Column view. $A\mathbf{x}$ is a linear combination of the **columns** of A:

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n \in \mathbb{R}^m.$$

The weights are the entries of $\mathbf{x}$.

Example. $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$, $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$:

$$A\mathbf{x} = x_1 \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix} + x_2 \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}.$$

3.3 Row vector times matrix — rows of A

Setup. $A \in \mathbb{R}^{m\times n}$, $\mathbf{x} \in \mathbb{R}^n$ (column), so $A\mathbf{x} \in \mathbb{R}^m$.

Row view (transpose both sides). $(A\mathbf{x})^\top = \mathbf{x}^\top A^\top$:

$$\mathbf{x}^\top A^\top = x_1\,\mathbf{a}_1^\top + x_2\,\mathbf{a}_2^\top + \cdots + x_n\,\mathbf{a}_n^\top \in \mathbb{R}^{1\times m}.$$

Dimensions: $(1\times n)(n\times m) = (1\times m)$. **Note:** $\mathbf{x}^\top A$ does **not** match when $A \in \mathbb{R}^{m\times n}$ with $m \neq n$.

Example. From the previous slide,

$$\mathbf{x}^\top = (x_1, x_2), \qquad A^\top = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}.$$

Then

$$\mathbf{x}^\top A^\top = x_1(1,3,5) + x_2(2,4,6).$$

3.3 Two views of the same product

Same numbers, two interpretations. If $A \in \mathbb{R}^{m\times n}$ and $\mathbf{x} \in \mathbb{R}^n$:

- **Column view:** $A\mathbf{x} = \sum_{j=1}^{n} x_j\, \mathbf{a}_j \in \mathbb{R}^m$.
- **Entry view:** $(A\mathbf{x})_i = \mathbf{r}_i^\top A\mathbf{x} = \sum_{j=1}^{n} a_{ij}x_j$ (dot product of row i of A with $\mathbf{x}$), where $\mathbf{r}_i \in \mathbb{R}^m$ has i-th entry 1 and all other entries 0 ($\mathbf{r}_i = \mathbf{e}_i$).

Transpose link. $(A\mathbf{x})^\top = \mathbf{x}^\top A^\top$: a **row vector** on the left combines **rows of** $A^\top$ (i.e. transposes of columns of A).

3.3 Matrix multiplication AB — example

Example. $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$.

Column view (both columns of AB).

$$(AB)\mathbf{e}_1 = A\begin{pmatrix} 5 \\ 7 \end{pmatrix} = 5\begin{pmatrix} 1 \\ 3 \end{pmatrix} + 7\begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 19 \\ 43 \end{pmatrix},$$

$$(AB)\mathbf{e}_2 = A\begin{pmatrix} 6 \\ 8 \end{pmatrix} = 6\begin{pmatrix} 1 \\ 3 \end{pmatrix} + 8\begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 22 \\ 50 \end{pmatrix}.$$

So $AB = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}$.

Row view (practice). Find rows 1 and 2 of AB using $(1, 2)B$ and $(3, 4)B$.

Note (skip in class). $(1, 2)B = (19, 22)$ and $(3, 4)B = (43, 50)$ — same matrix as above.

3.3 Matrix multiplication AB

Setup. $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times p}$, so $AB \in \mathbb{R}^{m\times p}$.

Column view. Column j of AB is A times column j of B:

$$(AB)\,\mathbf{e}_j = A(B\mathbf{e}_j) = A\mathbf{b}_j = b_{1j}\mathbf{a}_1 + b_{2j}\mathbf{a}_2 + \cdots + b_{nj}\mathbf{a}_n.$$

Each column of AB is a linear combination of **columns of** A (weights from column j of B).

Row view. Row i of AB is row i of A times B:

$$\mathbf{r}_i^{\mathsf{T}}(AB) = (\mathbf{r}_i^{\mathsf{T}}A)B = a_{i1}\mathbf{s}_1^{\mathsf{T}} + \cdots + a_{in}\mathbf{s}_n^{\mathsf{T}},$$

where $\mathbf{r}_i \in \mathbb{R}^m$ has i-th entry 1 and all other entries 0 ($\mathbf{r}_i = \mathbf{e}_i$), so $\mathbf{r}_i^{\mathsf{T}}$ selects row i; and $\mathbf{s}_k^{\mathsf{T}}$ is row k of B: a linear combination of **rows of** B.

3.3 Basis

Idea. A **basis** of $\mathbb{R}^n$ is a small set of vectors from which every vector in $\mathbb{R}^n$ can be built as a **linear combination** in exactly one way.

Standard basis of $\mathbb{R}^n$. For $j = 1, \ldots, n$, define $\mathbf{e}_j$ to have a 1 in entry j and 0 elsewhere:

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \ldots, \quad \mathbf{e}_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

Coordinates. Every $\mathbf{x} \in \mathbb{R}^n$ can be written uniquely as

$$\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n,$$

where $x_1, \ldots, x_n$ are the **coordinates** (entries) of $\mathbf{x}$.

3.3 Basis — example ($\mathbb{R}^2$)

Key point. A basis is **not unique**, but for a **fixed** basis every vector has a **unique** representation as a linear combination.

Two bases of $\mathbb{R}^2$. Standard: $\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Alternative: $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Same vector, two representations. For $\mathbf{b} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$:

$$\mathbf{b} = 2\mathbf{e}_1 + 3\mathbf{e}_2 \qquad \text{and} \qquad \mathbf{b} = (-1)\mathbf{v}_1 + 3\mathbf{v}_2.$$

Different bases give different weights $x_1, \ldots, x_n$, but within one basis the weights are **unique**.

Requirement for a basis $\{\mathbf{a}_1, \ldots, \mathbf{a}_n\}$. Every $\mathbf{b} \in \mathbb{R}^n$ can be written as $\mathbf{b} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$ in **exactly one way**.

3.3 Basis — existence and uniqueness

Requirement for a basis $\{\mathbf{a}_1, \ldots, \mathbf{a}_n\}$. Every $\mathbf{b} \in \mathbb{R}^n$ can be written as $\mathbf{b} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$ in **exactly one way**.

Two parts.

- **Existence:** every $\mathbf{b} \in \mathbb{R}^n$ can be written as some linear combination of $\mathbf{a}_1, \ldots, \mathbf{a}_n$.
- **Uniqueness:** there is **only one** choice of weights $x_1, \ldots, x_n$.

For $A\mathbf{x} = \mathbf{b}$ with $A \in \mathbb{R}^{m\times n}$. Then $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{b} \in \mathbb{R}^m$, and $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$ with $\mathbf{a}_j \in \mathbb{R}^m$ gives

$$A\mathbf{x} = \mathbf{b} \quad \Longleftrightarrow \quad \mathbf{b} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n.$$

3.3 Existence — column space

Setup. $A \in \mathbb{R}^{m \times n}$, $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{b} \in \mathbb{R}^m$, $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$ with $\mathbf{a}_j \in \mathbb{R}^m$.

Column space.

$$\mathrm{Col}(A) = \mathrm{span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\} \subseteq \mathbb{R}^m.$$

Existence of a solution. $A\mathbf{x} = \mathbf{b}$ has **at least one** solution $\iff$ $\mathbf{b} \in \mathrm{Col}(A)$.

Meaning. $\mathbf{b} \in \mathrm{Col}(A)$ means $\mathbf{b} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$ for some $\mathbf{x}$ — the **existence** part of the basis requirement, but only for this one $\mathbf{b} \in \mathbb{R}^m$ (and only when $\mathrm{Col}(A) = \mathbb{R}^m$ does it hold for **all b**).

3.3 Uniqueness — linearly independent columns

Setup. $A \in \mathbb{R}^{m\times n}$, $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{b} \in \mathbb{R}^m$.

Linearly independent. Vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are **linearly independent** if

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k = \mathbf{0} \quad \Longrightarrow \quad c_1 = c_2 = \cdots = c_k = 0.$$

Linearly dependent: not independent — some nontrivial combination equals $\mathbf{0}$.

Uniqueness of a solution. If $A\mathbf{x} = \mathbf{b}$ has a solution, it is **unique** iff the columns $\mathbf{a}_1, \dots, \mathbf{a}_n$ are **linearly independent**.

3.3 Linear independence $\Leftrightarrow$ unique solution

Claim. Fix $\mathbf{b} \in \mathbb{R}^m$. If $A\mathbf{x} = \mathbf{b}$ has **at least one** solution, then it has a **unique** solution iff $\mathbf{a}_1, \dots, \mathbf{a}_n$ are linearly independent.

($\Rightarrow$) Independent $\Rightarrow$ unique. If $A\mathbf{x} = \mathbf{b}$ and $A\mathbf{y} = \mathbf{b}$, then $A(\mathbf{x} - \mathbf{y}) = \mathbf{0}$, i.e.

$$(x_1 - y_1)\mathbf{a}_1 + \cdots + (x_n - y_n)\mathbf{a}_n = \mathbf{0}.$$

Independence gives $x_j - y_j = 0$ for all j, so $\mathbf{x} = \mathbf{y}$.

($\Leftarrow$) Dependent $\Rightarrow$ not unique. If the columns are dependent, some $\mathbf{d} \neq \mathbf{0}$ satisfies $A\mathbf{d} = \mathbf{0}$. If $\mathbf{x}$ solves $A\mathbf{x} = \mathbf{b}$, then so does $\mathbf{x} + \mathbf{d}$:

$$A(\mathbf{x} + \mathbf{d}) = A\mathbf{x} + A\mathbf{d} = \mathbf{b} + \mathbf{0} = \mathbf{b},$$

and $\mathbf{x} + \mathbf{d} \neq \mathbf{x}$. So the solution is **not** unique.

3.3 Linear independence — examples ($\mathbb{R}^2$)

Independent (bases).

- Standard: $\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.
- Alternative: $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

For each pair, $c_1\mathbf{w}_1 + c_2\mathbf{w}_2 = \mathbf{0}$ implies $c_1 = c_2 = 0$.

Dependent (too many vectors). The same $\mathbf{e}_1, \mathbf{e}_2$, and $\mathbf{u} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$:

$$\mathbf{u} = \mathbf{e}_1 + \mathbf{e}_2 \quad \Longleftrightarrow \quad \mathbf{e}_1 + \mathbf{e}_2 - \mathbf{u} = \mathbf{0}.$$

The weights 1, 1, and -1 are not all zero, so these three vectors are **linearly dependent**. A basis of $\mathbb{R}^2$ has exactly 2 vectors, not 3.

3.3 Linear dependence — columns example

Example (columns of A). $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$, columns $\mathbf{a}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $\mathbf{a}_2 = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$.

Same direction. $\mathbf{a}_2 = 2\mathbf{a}_1$, so the two columns lie on one line through the origin.

Nontrivial combination to 0.

$$2\mathbf{a}_1 - \mathbf{a}_2 = \mathbf{0} \quad \Longleftrightarrow \quad 2\begin{pmatrix} 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

The weights 2 and -1 are not both zero, so $\mathbf{a}_1, \mathbf{a}_2$ are **linearly dependent** (only one independent direction in $\mathrm{Col}(A)$).

3.3 Solving $A\mathbf{x} = \mathbf{b}$ — example

Example. $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$, so $\mathrm{Col}(A) = \mathrm{span}\{\begin{pmatrix} 1 \\ 2 \end{pmatrix}\}$ (a line in $\mathbb{R}^2$).

Solvable. $\mathbf{b} = \begin{pmatrix} 3 \\ 6 \end{pmatrix} = 3\begin{pmatrix} 1 \\ 2 \end{pmatrix}$, so $A\mathbf{x} = \mathbf{b}$ has solutions (e.g. $\mathbf{x} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$; not unique).

Not solvable (existence fails). $\mathbf{b} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \notin \mathrm{Col}(A)$, so $A\mathbf{x} = \mathbf{b}$ has **no** solution.

Dependent columns (uniqueness fails). $\mathbf{a}_2 = 2\mathbf{a}_1$, so the columns are **linearly dependent**. When a solution exists, it is not unique (e.g. both $\mathbf{x} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$ and $\mathbf{x} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ give $\mathbf{b} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$).

3.3 Full basis requirement — only when $m = n$

Goal. When does **every** $\mathbf{b} \in \mathbb{R}^m$ satisfy $\mathbf{b} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$ in **exactly one way**?

Claim. For $A \in \mathbb{R}^{m\times n}$, this happens **only if** $m = n$.

Proof sketch.

- **Uniqueness** $\Rightarrow$ columns independent. In $\mathbb{R}^m$ at most m vectors can be independent, so $n \leq m$.
- **Existence for all b** $\Rightarrow \mathrm{Col}(A) = \mathbb{R}^m$. With only n columns, $\dim(\mathrm{Col}(A)) \leq n$, so $n \geq m$.
- Therefore $m = n$.

If $n < m$, columns cannot span $\mathbb{R}^m$; if $n > m$, n columns in $\mathbb{R}^m$ cannot be independent.

3.3 Square case — basis = existence + uniqueness

Now $A \in \mathbb{R}^{n\times n}$, $\mathbf{b} \in \mathbb{R}^n$. Requirement for a basis $\{\mathbf{a}_1, \ldots, \mathbf{a}_n\}$: every $\mathbf{b} \in \mathbb{R}^n$ is $\mathbf{b} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$ in **exactly one way**.

Equivalent to $A\mathbf{x} = \mathbf{b}$.

$$A\mathbf{x} = \mathbf{b} \quad \Longleftrightarrow \quad \mathbf{b} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n.$$

Columns form a basis of $\mathbb{R}^n$ $\Longleftrightarrow$ existence and uniqueness for **every $\mathbf{b}$** $\Longleftrightarrow$ $A\mathbf{x} = \mathbf{b}$ has a **unique** solution for every $\mathbf{b} \in \mathbb{R}^n$.

3.4 Two views of a matrix A

View 1 — $\mathbf{x}$ as weights. Write $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$. Then

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

The entries of $\mathbf{x}$ are the **weights** on the columns of A.

View 2 — A as a linear transformation. Define $T : \mathbb{R}^n \to \mathbb{R}^m$ by $T(\mathbf{x}) = A\mathbf{x}$. The matrix A sends each input vector $\mathbf{x}$ to an output vector in $\mathbb{R}^m$; for square A, $|\det(A)|$ is the area-scaling factor (Section 3.5).

3.4 Linear transformation — definition

Linear transformation. A map $T : \mathbb{R}^n \to \mathbb{R}^m$ is **linear** if for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and all scalars $c \in \mathbb{R}$:

$$T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y}), \qquad T(c\mathbf{x}) = cT(\mathbf{x}).$$

Equivalently, for all $\mathbf{x}, \mathbf{y}$ and c:

$$T(c\mathbf{x} + \mathbf{y}) = cT(\mathbf{x}) + T(\mathbf{y}).$$

Examples. Rotation, reflection, projection, and shear in the plane are linear. A map with a constant term, e.g. $T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$ with $\mathbf{b} \neq \mathbf{0}$, is **not** linear.

3.4 Equivalence: linear T and matrix A

Take $\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, so any $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ is

$$\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2.$$

If you know $T(\mathbf{e}_1)$ and $T(\mathbf{e}_2)$, then by linearity

$$T(\mathbf{v}) = T(v_1\mathbf{e}_1 + v_2\mathbf{e}_2) = v_1 T(\mathbf{e}_1) + v_2 T(\mathbf{e}_2).$$

Stack those outputs as columns of A:

$$A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) \end{bmatrix} \qquad \Longrightarrow \qquad T(\mathbf{v}) = A\mathbf{v}.$$

Every linear T is multiplication by this A; every matrix A defines a linear $T(\mathbf{x}) = A\mathbf{x}$. Same object, two languages.

3.4 Every linear map is a matrix

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be linear, and write $\mathbf{x} = \sum_{j=1}^n x_j \mathbf{e}_j$. Then

$$T(\mathbf{x}) = T\Big(\sum_{j=1}^{n} x_j \mathbf{e}_j\Big) = \sum_{j=1}^{n} x_j T(\mathbf{e}_j).$$

Set

$$A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{bmatrix}.$$

Column j of A is $T(\mathbf{e}_j)$, so the same sum is $A\mathbf{x}$:

$$T(\mathbf{x}) = A\mathbf{x}.$$

This A is unique: column j must be where $\mathbf{e}_j$ is sent.

3.4 Why $(AB)^\mathsf{T} = B^\mathsf{T} A^\mathsf{T}$

B sends $\mathbb{R}^p \to \mathbb{R}^n$, then A sends $\mathbb{R}^n \to \mathbb{R}^m$, so AB sends $\mathbb{R}^p \to \mathbb{R}^m$.

Transpose reverses the arrows: $(AB)^\mathsf{T}$ sends $\mathbb{R}^m \to \mathbb{R}^p$.

To go back, undo A first then B: A^T sends $\mathbb{R}^m \to \mathbb{R}^n$, then B^T sends $\mathbb{R}^n \to \mathbb{R}^p$. That composition is $B^\mathsf{T} A^\mathsf{T}$.

$A^\mathsf{T} B^\mathsf{T}$ would need $\mathbb{R}^p = \mathbb{R}^m$ to even compose. So

$$(AB)^\mathsf{T} = B^\mathsf{T} A^\mathsf{T}.$$

3.4 Decompose, transform, combine — notation

Recipe. For $\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2$ and linear T:

1. Decompose the input:

$$\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2.$$

2. Transform each piece (use known values of T on $\mathbf{e}_1, \mathbf{e}_2$):

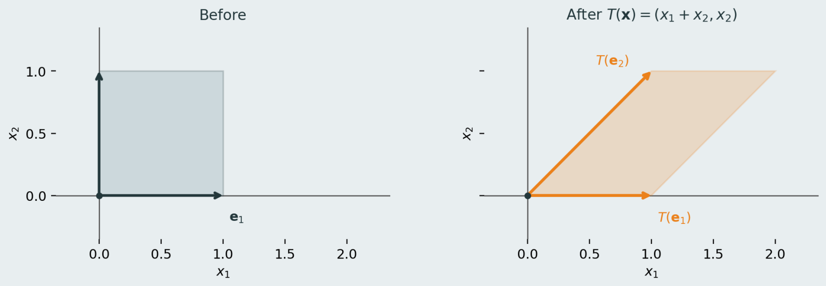
$$v_1 T(\mathbf{e}_1), \qquad v_2 T(\mathbf{e}_2).$$

3. Combine the outputs:

$$T(\mathbf{v}) = v_1 T(\mathbf{e}_1) + v_2 T(\mathbf{e}_2) = A\mathbf{v}, \qquad A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) \end{bmatrix}.$$

3.4 Example — shear

Map. Horizontal shear: tilt the unit square into a parallelogram, $(x_1, x_2) \mapsto (x_1 + x_2, x_2)$.



3.4 Example — shear (matrix)

Find the matrix. Apply T to $\mathbf{e}_1$ and $\mathbf{e}_2$, then stack the outputs as columns:

$$T(\mathbf{e}_1) = \mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad T(\mathbf{e}_2) = \mathbf{e}_1 + \mathbf{e}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \Longrightarrow \quad A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Check: $A\mathbf{e}_1 = \mathbf{e}_1$ (bottom edge fixed); $A\mathbf{e}_2$ tilts the vertical side.

3.4 Example — shear (decompose, transform, combine)

Take $\mathbf{v} = 2\mathbf{e}_1 + 3\mathbf{e}_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

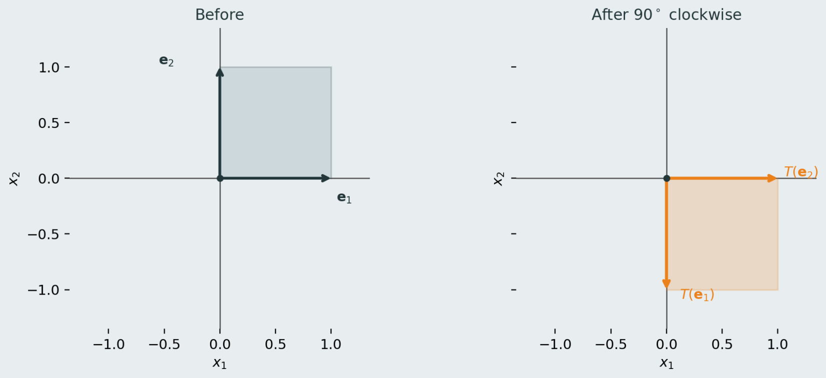
1. Decompose: $\mathbf{v} = 2\mathbf{e}_1 + 3\mathbf{e}_2$.

2. Transform: $2T(\mathbf{e}_1) = 2\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad 3T(\mathbf{e}_2) = 3\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$.

3. Combine:

$$T(\mathbf{v}) = 2T(\mathbf{e}_1) + 3T(\mathbf{e}_2) = \begin{pmatrix} 2 \\ 0 \end{pmatrix} + \begin{pmatrix} 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \end{pmatrix} = A\mathbf{v}.$$

Map. Rotate every vector 90° clockwise: $(x_1, x_2) \mapsto (x_2, -x_1)$.



3.4 Example — rotation (matrix)

Find the matrix. Apply T to $\mathbf{e}_1$ and $\mathbf{e}_2$, then stack the outputs as columns:

$$T(\mathbf{e}_1) = \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \qquad T(\mathbf{e}_2) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \Longrightarrow \quad A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Check: $A\mathbf{e}_1$ points down; $A\mathbf{e}_2$ points right (see previous graph).

3.4 Example — rotation (decompose, transform, combine)

Same recipe with $\mathbf{v} = 2\mathbf{e}_1 + 3\mathbf{e}_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

1. Decompose: $\mathbf{v} = 2\mathbf{e}_1 + 3\mathbf{e}_2$.

2. Transform: $2T(\mathbf{e}_1) = 2\begin{pmatrix} 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}, \quad 3T(\mathbf{e}_2) = 3\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$.

3. Combine:

$$T(\mathbf{v}) = 2T(\mathbf{e}_1) + 3T(\mathbf{e}_2) = \begin{pmatrix} 0 \\ -2 \end{pmatrix} + \begin{pmatrix} 3 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} = A\mathbf{v}.$$

3.5 Determinant — area scaling (2×2)

Setup. $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, columns $\mathbf{a}_1 = \begin{pmatrix} a \\ c \end{pmatrix}$, $\mathbf{a}_2 = \begin{pmatrix} b \\ d \end{pmatrix}$. View A as the linear map $T(\mathbf{x}) = A\mathbf{x}$ (shear, rotation, etc. on previous slides).

Where the standard basis goes.

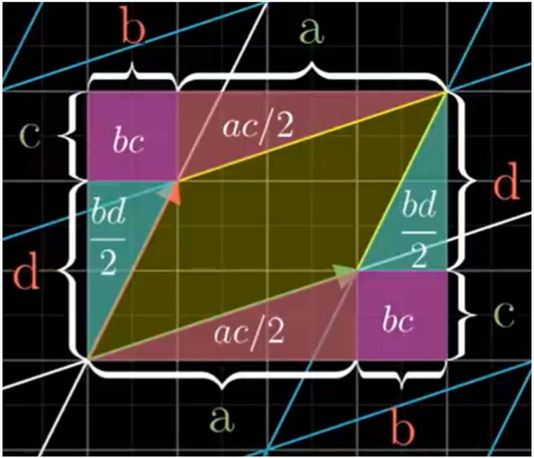
$$A\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ c \end{pmatrix} = \mathbf{a}_1, \qquad A\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} b \\ d \end{pmatrix} = \mathbf{a}_2.$$

Area conversion factor. The unit square (spanned by $\mathbf{e}_1$ and $\mathbf{e}_2$, area $= 1$) is mapped by A to the parallelogram spanned by $\mathbf{a}_1$ and $\mathbf{a}_2$. The **determinant** is exactly the (signed) area of that parallelogram:

$$\det(A) = ad - bc.$$

So $\det(A)$ is the **area conversion factor**: every region's area is multiplied by $|\det(A)|$ under $\mathbf{x} \mapsto A\mathbf{x}$.

3.5 Determinant — why $ad - bc$ is the area



Box area $(a + b)(c + d)$ minus purple (bc), red ($ac/2$), teal ($bd/2$) pieces leaves yellow parallelogram area

$$ad - bc = \det(A).$$

3.5 Determinant $= 0$ — example

Example. $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, so $a = b = c = d = 1$ and

$$\det(A) = ad - bc = 1 \cdot 1 - 1 \cdot 1 = 0.$$

Standard basis collapses to a line.

$$A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad A \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Both $\mathbf{e}_1$ and $\mathbf{e}_2$ are sent to the **same** point on one line through the origin. The unit square (area 1) is squashed to a **1-dimensional** set — area 0.

So (square case):

$$\det(A) = 0 \iff \text{columns linearly dependent.}$$

Here $\mathbf{a}_2 = \mathbf{a}_1$, so the columns are dependent.

3.5 Null space — directions squeezed to 0

Setup ($A \in \mathbb{R}^{2\times 2}$). View A as a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(\mathbf{x}) = A\mathbf{x}$.

Null space / kernel (subspace of $\mathbb{R}^2$):

$$\mathrm{Null}(A) = \ker(T) = \{\mathbf{x} \in \mathbb{R}^2 : \; A\mathbf{x} = \mathbf{0}\}.$$

Vectors in $\mathrm{Null}(A)$ are mapped to $\mathbf{0}$ in the output.

Intuition. If $\mathbf{x} \neq \mathbf{0}$ lies in $\mathrm{Null}(A)$, the whole line $\mathrm{span}\{\mathbf{x}\} \subset \mathbb{R}^2$ is sent to the single point $\mathbf{0} \in \mathbb{R}^2$ — that **input direction is squeezed** (lost in the output).

Example ($\det A = 0$). $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$: $A\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \mathbf{0}$, so $\mathrm{Null}(A) = \mathrm{span}\left\{\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right\}$ — one direction squeezed.

Next: how many directions squeeze vs. persist?

3.5 Rank–nullity — squeezed vs. persist (2×2)

Two fates ($A \in \mathbb{R}^{2\times 2}$). Under $T(\mathbf{x}) = A\mathbf{x}$, each input direction either:

1. Squeezed to 0 $\ker(T) = \mathrm{Null}(A)$ (previous slide); dimension $= \mathrm{nullity}(A)$.

2. Persist in the output $\mathrm{Im}(T) = \mathrm{Col}(A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^2\}$; dimension $= \mathrm{rank}(A)$ — independent directions that **survive**.

Rank–nullity. $\mathrm{rank}(A) + \mathrm{nullity}(A) = 2$.

Example ($\det A = 0$). Collapse matrix from previous slides: nullity $= 1$, rank $= 1$ (outputs fill $\mathrm{span}\{\begin{pmatrix}1\\1\end{pmatrix}\}$).

Example ($\det A \neq 0$). Shear or rotation (Section 3.4): nullity $= 0$, rank $= 2$.

3.5 After a squeeze — what survives? (2×2)

Recall. $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ maps the whole plane onto one output line through $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Direction $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is squeezed to $\mathbf{0}$.

Key idea. Split $\mathbf{v}$ into a squeezed part (forgotten) and a surviving part along $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ (remembered). So $A\mathbf{v}$ depends only on the surviving part.

Example.

$$\mathbf{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

The squeezed part is lost, so $A\mathbf{v} = 2A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 2 \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$.

Next: the same squeeze/persist picture in general $m \times n$.

3.5 Rank–nullity — general case for interest only

Setup. Let $A \in \mathbb{R}^{m\times n}$, $T : \mathbb{R}^n \to \mathbb{R}^m$, $T(\mathbf{x}) = A\mathbf{x}$.

Kernel and image. $\ker(T) = \mathrm{Null}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\} \subseteq \mathbb{R}^n$;
$\mathrm{Im}(T) = \mathrm{Col}(A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\} \subseteq \mathbb{R}^m$.

Rank and nullity. $\mathrm{rank}(A) = \dim(\mathrm{Im}\ T)$, $\quad \mathrm{nullity}(A) = \dim(\ker T)$.

Rank–nullity theorem.

$$\mathrm{rank}(A) + \mathrm{nullity}(A) = n \qquad (n \text{ columns} = \dim(\mathbb{R}^n)).$$

Same intuition. Of the n domain dimensions, $\mathrm{nullity}(A)$ are **squeezed** to $\mathbf{0} \in \mathbb{R}^m$; $\mathrm{rank}(A)$ independent directions **persist** in $\mathrm{Im}(T) \subseteq \mathbb{R}^m$. When $n = m$, $\det(A) \neq 0 \Leftrightarrow \mathrm{nullity} = 0 \Leftrightarrow \mathrm{rank} = n$ (same as invertible / basis on the next slide).

3.5 Invertible matrices — column picture

Recall ($A \in \mathbb{R}^{n\times n}$). $A\mathbf{x} = \mathbf{b} \iff \mathbf{b} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$ (§3.3).

Basis = two requirements (§3.3). Columns $\mathbf{a}_1, \ldots, \mathbf{a}_n$ form a **basis** of $\mathbb{R}^n$ iff:

(1) Uniqueness: at most one choice of weights $x_1, \ldots, x_n$ (columns **linearly independent**).

(2) Solvability: **every $\mathbf{b} \in \mathbb{R}^n$** *is* such a combination ($\mathrm{Col}(A) = \mathbb{R}^n$).

For square A, (1)⇔(2); together they give a **unique** solution for **every b**, and A^{-1} exists.

Geometry.

- A squeeze ($\det(A) = 0$): some direction goes to $\mathbf{0}$ — not a basis, not invertible.
- No squeeze ($\det(A) \neq 0$): nullity $= 0$ — a basis, invertible.

3.5 Invertible matrices — equivalent conditions

Let $A \in \mathbb{R}^{n\times n}$. All eight statements are **equivalent**.

I. Column picture (§3.3)

1. Columns **linearly independent** — **uniqueness**.
2. $\mathrm{Col}(A) = \mathbb{R}^n$ — **solvability for all b**.
3. Columns form a **basis** — (1) and (2) [square: (1)⇔(2)].

II. Solving $A\mathbf{x} = \mathbf{b}$

4. $A\mathbf{x} = \mathbf{0}$ only $\mathbf{x} = \mathbf{0}$.
5. For every $\mathbf{b}$, **unique** solution $\mathbf{x} = A^{-1}\mathbf{b}$.

III. Matrix / geometry

6. A **invertible** ($AA^{-1} = I_n$).
7. $\det(A) \neq 0$ (no squeeze).
8. Rows form a **basis** of $\mathbb{R}^n$.

3.5 Why (8): rows form a basis

Goal. (2) $\Rightarrow$ (8): columns of A span $\mathbb{R}^n$ $\Rightarrow$ rows of A form a basis.

Observation. Rows of A = columns of A^{T}. So (8) is just "columns of A^{T} form a basis."

- (2) $\Rightarrow$ A invertible: $AA^{-1} = I_n$.
- Transpose both sides: $(A^{-1})^{\mathsf{T}} A^{\mathsf{T}} = I_n$, so A^{T} is invertible too.
- Invertible $\Rightarrow$ columns of A^{T} form a basis.
- Those columns are the rows of A. That is (8).

3.5 Transpose and inverse — rules we use

Transpose (always defined).

$$(A^{\mathsf{T}})^{\mathsf{T}} = A, \qquad (AB)^{\mathsf{T}} = B^{\mathsf{T}} A^{\mathsf{T}}, \qquad (A\mathbf{x})^{\mathsf{T}} = \mathbf{x}^{\mathsf{T}} A^{\mathsf{T}}.$$

Inverse (definition). For square $A \in \mathbb{R}^{n\times n}$, A^{-1} satisfies $AA^{-1} = A^{-1}A = I_n$. It exists iff any condition on the equivalent-conditions list holds (e.g. $\det(A) \neq 0$). Then $A\mathbf{x} = \mathbf{b}$ has unique solution $\mathbf{x} = A^{-1}\mathbf{b}$.

Inverse rules (both factors invertible).

$$(AB)^{-1} = B^{-1}A^{-1}, \qquad (A^{-1})^{-1} = A, \qquad (A^{-1})^{\mathsf{T}} = (A^{\mathsf{T}})^{-1}.$$

Proofs: appendix (**for interest only**).

3.6 Dot product — algebraic and geometric

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$:

Algebraic definition.

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{n} u_i v_i .$$

Geometric definition. Let θ be the angle between $\mathbf{u}$ and $\mathbf{v}$. Then

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \, \|\mathbf{v}\| \cos\theta .$$

Equivalent. These two definitions always give the **same number**.

Video (intuition and proof). https://www.youtube.com/watch?v=LyGKycYT2v0

3.6 Linear regression — $\mathbf{y}$ and X

Data (n workers). Worker i has income y_i, age age_i, education educ_i.

Response vector.

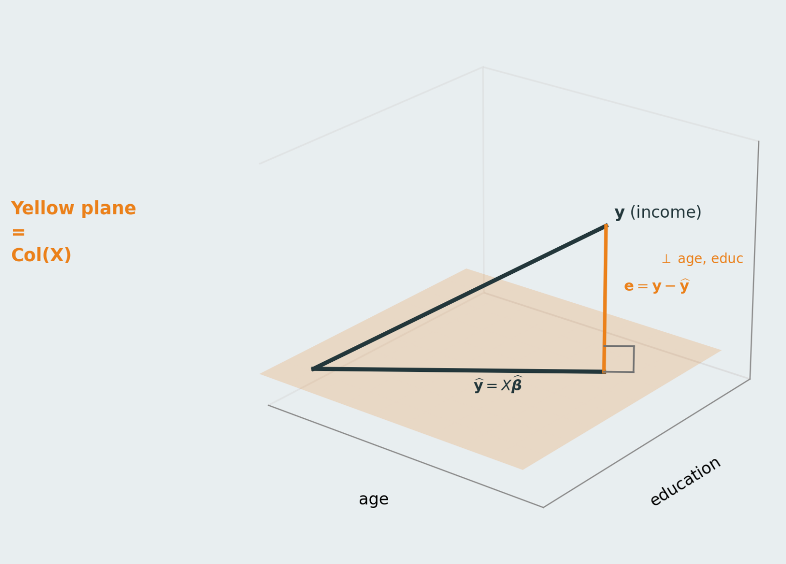
$$\mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^n.$$

Regressor matrix (two columns: age, educ).

$$X = \begin{pmatrix} \text{age}_1 & \text{educ}_1 \\ \vdots & \vdots \\ \text{age}_n & \text{educ}_n \end{pmatrix} \in \mathbb{R}^{n\times 2}, \qquad \boldsymbol{\beta} = \begin{pmatrix} \beta_{\text{age}} \\ \beta_{\text{educ}} \end{pmatrix}, \qquad \mathbf{y} \approx X\boldsymbol{\beta}.$$

Column j of X is the vector of worker i values for regressor j.

Schematic next ($n = 3$): age and education are the two axes; yellow plane $= \text{Col}(X)$; $\mathbf{e}$ is orthogonal to every column of X.



Yellow plane
=
Col(X)
y (income)
⊥ age, educ
$\mathbf{e} = \mathbf{y} - \widehat{\mathbf{y}}$
$\widehat{\mathbf{y}} = X\widehat{\boldsymbol{\beta}}$
age
education

3.6 OLS fit — e orthogonal to every column of X

From the picture ($n = 3$). Yellow plane $=$ Col(X); fitted $\widehat{\mathbf{y}} = X\widehat{\boldsymbol{\beta}}$ lies in it. Residual $\mathbf{e} = \mathbf{y} - X\widehat{\boldsymbol{\beta}}$ is perpendicular to the plane.

Orthogonal to every column of X. Let $\mathbf{x}_1 =$ age column, $\mathbf{x}_2 =$ educ column. Then

$$\mathbf{x}_1 \cdot \mathbf{e} = 0, \qquad \mathbf{x}_2 \cdot \mathbf{e} = 0 \qquad \Longleftrightarrow \qquad X^\top \mathbf{e} = \mathbf{0} \qquad \Longleftrightarrow \qquad X^\top\big(\mathbf{y} - X\widehat{\boldsymbol{\beta}}\big) = \mathbf{0}.$$

These **normal equations** characterise $\widehat{\boldsymbol{\beta}}$.

Solve for $\widehat{\boldsymbol{\beta}}$. $X^\top X\widehat{\boldsymbol{\beta}} = X^\top \mathbf{y}$. If the columns of X are independent (full column rank), then

$$\widehat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}.$$

(Since $\widehat{\mathbf{y}} \in$ Col(X), also $\widehat{\mathbf{y}} \cdot \mathbf{e} = 0$.)

3.6 OLS fit — same as least squares

Pythagoras in $\mathbb{R}^n$. Fix $\widehat{\boldsymbol{\beta}}$ with $\mathbf{e} = \mathbf{y} - X\widehat{\boldsymbol{\beta}}$. For any $\boldsymbol{\beta}$,

$$\mathbf{y} - X\boldsymbol{\beta} = \underbrace{\mathbf{y} - X\widehat{\boldsymbol{\beta}}}_{\mathbf{e}} + \underbrace{X(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta})}_{\in \mathrm{Col}(X)}.$$

The two pieces are orthogonal, so

$$\|\mathbf{y} - X\boldsymbol{\beta}\|^2 = \|\mathbf{y} - X\widehat{\boldsymbol{\beta}}\|^2 + \|X(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 \geq \|\mathbf{y} - X\widehat{\boldsymbol{\beta}}\|^2.$$

Conclusion. The smallest squared error occurs when $X(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \mathbf{0}$, i.e.
$\widehat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\mathbf{y} - X\boldsymbol{\beta}\|^2 = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} \big(y_i - \beta_{\text{age}}\, \text{age}_i - \beta_{\text{educ}}\, \text{educ}_i\big)^2.$

3.6 The projection matrix

Projection matrix P_X (hat matrix). If $X \in \mathbb{R}^{n \times k}$ has full column rank, define

$$P_X = X(X^\top X)^{-1} X^\top .$$

Then $\widehat{\mathbf{y}} = P_X \mathbf{y}$.

Annihilator matrix M_X (residual maker matrix). Define $M_X = I - P_X$. Then $\mathbf{e} = M_X \mathbf{y}$; and $M_X X = \mathbf{0}$ (M_X **annihilates** Col(X)).

Useful facts. $P_X^\top = P_X$, $P_X^2 = P_X$, $M_X X = \mathbf{0}$, and $M_X^\top = M_X$, $M_X^2 = M_X$.

Why $P_X^\top = P_X$ and $P_X^2 = P_X$: appendix (**for interest only**).

3.6 Frisch–Waugh–Lovell (FWL) theorem for interest only

Partition regressors. Write $X = [X_1 \mid X_2]$ (e.g. age $= X_1$, educ $= X_2$). Let P_{X_2} be the **projection matrix** onto $\mathrm{Col}(X_2)$ and $M_{X_2} = I - P_{X_2}$ the **annihilator matrix** (residual maker matrix; write P_2, M_2 for short).

Claim. The OLS coefficient on X_1 in $\mathbf{y} \sim X_1 + X_2$ equals the OLS coefficient in

$$M_2\mathbf{y} \;\sim\; M_2X_1$$

(M_2 removes the part of each vector explained by educ).

Reading. "Controlling for educ" $=$ apply the annihilator M_2 to both income and age, then regress.

Proof idea. Write $\mathbf{y} = X_1\widehat{\boldsymbol{\beta}}_1 + X_2\widehat{\boldsymbol{\beta}}_2 + \mathbf{e}$ with $X^\mathsf{T}\mathbf{e} = \mathbf{0}$. Since $M_2X_2 = \mathbf{0}$ and $M_2\mathbf{e} = \mathbf{e}$, get $M_2\mathbf{y} = M_2X_1\widehat{\boldsymbol{\beta}}_1 + \mathbf{e}$; also $(M_2X_1)^\mathsf{T}\mathbf{e} = X_1^\mathsf{T}M_2\mathbf{e} = X_1^\mathsf{T}\mathbf{e} = 0$, so $\widehat{\boldsymbol{\beta}}_1$ solves the partialled normal equations.

3.7 Which basis makes decompose–transform–combine easy?

Recall the recipe. If $\mathbf{v} = a\mathbf{x} + b\mathbf{y}$, then $T(\mathbf{v}) = aT(\mathbf{x}) + bT(\mathbf{y})$ — always true by linearity.

But step 2 can be hard. You must know $T(\mathbf{x})$ and $T(\mathbf{y})$. With the standard basis $(\mathbf{e}_1, \mathbf{e}_2)$, rotation sent $\mathbf{e}_1$ down and $\mathbf{e}_2$ right — both outputs are **new directions** to track.

A proper basis (for this map). Pick directions where T **only scales**, not rotates or shears:

$$T(\mathbf{v}) = \lambda\mathbf{v} \qquad \text{for some scalar } \lambda.$$

Then $T(a\mathbf{v}) = a\lambda\mathbf{v}$ — one number λ, same direction. These special directions are **eigenvectors**; λ is the **eigenvalue**.

3.7 Eigenvectors and eigenvalues — definition

Setup. Square matrix $A \in \mathbb{R}^{n \times n}$ (linear map $T(\mathbf{x}) = A\mathbf{x}$).

Eigenvector / eigenvalue. A nonzero vector $\mathbf{v}$ is an **eigenvector** of A if

$$A\mathbf{v} = \lambda \mathbf{v}$$

for some scalar λ (the **eigenvalue**). Geometrically: A stretches or flips $\mathbf{v}$, but keeps its line.

Proper basis. When A has enough independent eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$, decompose in **that** basis instead of the standard one.

If $A = A^\mathsf{T}$, eigenvalues are real (appendix) (**for interest only**).

3.7 Finding eigenvalues and eigenvectors

Traditional recipe (works in any dimension):

1. **Eigenvalues.** Solve the **characteristic equation**

$$\det(A - \lambda I) = 0.$$

2. **Eigenvectors.** For each eigenvalue λ, solve

$$(A - \lambda I)\mathbf{v} = \mathbf{0}.$$

   Nontrivial solutions are eigenvectors; the solution set is the **eigenspace** E_λ.

   Small n (e.g. 2×2): solve the equations by hand (Together exercise below).

   General / high n**:** row-reduce $(A - \lambda I)$ to **RREF** and read off pivot vs. free columns — appendix (**for interest only**).

3. **Repeated** λ**.** If λ has algebraic multiplicity > 1, $\dim E_\lambda$ equals the number of free variables in step 2.

Together: 2×2 by hand. Take home: $\mathbb{R}^4$ exercise below.

3.7 Exercise: Eigenvectors — 2×2 example Together

Exercise. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, find all eigenpairs and compute $A\mathbf{v}$ for $\mathbf{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$.

Solution (eigenvalues). Characteristic equation $\det(A - \lambda I) = 0$:

$$(2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3) = 0 \quad \Rightarrow \quad \lambda_1 = 3,\ \lambda_2 = 1.$$

Solution ($\lambda_1 = 3$). Solve $(A - 3I)\mathbf{v} = \mathbf{0}$:

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \mathbf{0} \quad \Rightarrow \quad -v_1 + v_2 = 0 \quad \Rightarrow \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Solution ($\lambda_2 = 1$). Solve $(A - I)\mathbf{v} = \mathbf{0}$:

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \mathbf{0} \quad \Rightarrow \quad v_1 + v_2 = 0 \quad \Rightarrow \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Check: $A\mathbf{v}_1 = 3\mathbf{v}_1$, $A\mathbf{v}_2 = 1\mathbf{v}_2$.

Compute $A\mathbf{v}$ for $\mathbf{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ (decompose–transform–combine).

Step 1 (decompose). Write $\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$:

$$c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \quad \Rightarrow \quad c_1 = 2,\ c_2 = 1.$$

Step 2 (transform). A scales each eigen-direction by its eigenvalue:

$$A\mathbf{v} = 2 \cdot 3\mathbf{v}_1 + 1 \cdot 1\mathbf{v}_2 = \begin{pmatrix} 7 \\ 5 \end{pmatrix}.$$

3.7 Exercise: Eigenvectors — 2×2 example (cont.) Together

Compute $A^{10}\mathbf{v}$ for $\mathbf{v} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} = 2\mathbf{v}_1 + \mathbf{v}_2$.

On an eigenvector: $A^{10}\mathbf{v}_1 = A^9(A\mathbf{v}_1) = A^9(\lambda_1\mathbf{v}_1) = \lambda_1 A^9\mathbf{v}_1 = \cdots = \lambda_1^{10}\mathbf{v}_1$ (and similarly $A^{10}\mathbf{v}_2 = \lambda_2^{10}\mathbf{v}_2$).

Same recipe: multiply each coefficient by λ_i^{10}:

$$A^{10}\mathbf{v} = 2 \cdot 3^{10}\mathbf{v}_1 + 1 \cdot 1^{10}\mathbf{v}_2.$$

Since $3^{10} = 59049$,

$$A^{10}\mathbf{v} = 118098 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 118099 \\ 118097 \end{pmatrix}.$$

The $\lambda_1 = 3$ term dominates; repeated multiplication amplifies $\mathbf{v}_1$.

3.7 Exercise: Eigenvectors in $\mathbb{R}^4$ (RREF) Take home

Exercise. For

$$A = \begin{pmatrix} 3 & 1 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 3 \end{pmatrix},$$

find all eigenvalues. For $\lambda = 2$, row-reduce $(A - 2I)$ to RREF and find a basis of E_2 (RREF: appendix).

Solution (eigenvalues). Two decoupled 2×2 blocks; each gives $(3 - \lambda)^2 - 1 = 0$.

$$\det(A - \lambda I) = \left((3 - \lambda)^2 - 1\right)^2 = (\lambda - 2)^2(\lambda - 4)^2.$$

So $\lambda = 2$ and $\lambda = 4$, each with algebraic multiplicity 2.

Solution ($\lambda = 2$). Row-reduce

$$A - 2I = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Pivot columns $1, 3 \Rightarrow v_1, v_3$ pivot; **free columns** $2, 4 \Rightarrow v_2, v_4$ free. Equations: $v_1 + v_2 = 0$, $v_3 + v_4 = 0$.

General solution / basis of E_2:

$$\mathbf{v} = v_2 \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + v_4 \begin{pmatrix} 0 \\ 0 \\ -1 \\ 1 \end{pmatrix}.$$

3.7 Eigenbasis — especially useful in high dimension

If $\mathbf{v}_1, \ldots, \mathbf{v}_n$ is an eigenbasis and $\mathbf{v} = \sum_i c_i \mathbf{v}_i$, then

$$A\mathbf{v} = \sum_i c_i \lambda_i \mathbf{v}_i, \qquad A^k\mathbf{v} = \sum_i c_i \lambda_i^k \mathbf{v}_i.$$

In an eigenbasis, A is just **separate scalings** — easy to describe and compute when n is large.

Not every matrix has a full real eigenbasis (e.g. plane rotation); appendix (**for interest only**). SVD/QR play a similar role when it does not.

3.7 Applications — PCA and Markov chains for interest only

for interest only You may need to understand some of these when you pursue macro, empirical economics, or IO.

Extremely useful beyond the 2×2 example. Many applied problems reduce to: find directions where a matrix **acts simply**.

- **PCA (principal components).** Eigenvectors of the **covariance matrix** are the main directions of variation in data; largest eigenvalues pick the most important components for dimension reduction (Section 3.8).
- **Markov transition matrices.** For a stochastic matrix P (rows sum to 1), the **long-run distribution** is an eigenvector with $\lambda = 1$; other eigenvalues control how fast $P^k \mathbf{v}$ converges to that steady state.

Pattern. Decompose a state or data vector in special directions; each direction evolves by a single number (λ_i).

3.7 Applications — ODEs for interest only

Ordinary differential equations (ODEs) — eigenvalues / eigenvectors show up in:

- Linear systems $\dot{\mathbf{x}} = A\mathbf{x}$
- Stability of equilibria and phase-plane dynamics
- Continuous-time macro and growth models
- Linearized dynamics around steady states
- Compartment models (e.g. epidemiology, population dynamics)

3.7 Decompositions — QR, eigen, SVD for interest only

Different decompositions, same spirit: rewrite a matrix so that the hard part is **diagonal**, **triangular**, or **orthogonal**.

- **QR decomposition** ($A = QR$). Q has orthonormal columns spanning $\mathrm{Col}(A)$; R is upper triangular. Used to pick **linearly independent columns**, solve least squares stably, and build regression numerically.
- **Spectral / eigen decomposition** ($A = V\Lambda V^{-1}$). When A has a full eigenbasis, V lists eigenvectors and Λ lists eigenvalues — matrix powers and dynamics become coefficient-wise.
- **Singular Value Decomposition (SVD)** ($A = U\Sigma V^{\top}$). Always exists; built from eigenproblems on $A^{\top}A$ (and $AA^{\top}$). Underlies PCA, low-rank approximation, and regularized regression.

Not required for Math Camp — names and connections only.

Self-study video. https://www.youtube.com/watch?v=PFDu9oVAE-g

Setup. Symmetric $A \in \mathbb{R}^{n \times n}$ (Hessians, covariance matrices, $A^\top A$).

Four types (via the **quadratic form** $\mathbf{v}^\top A\mathbf{v}$):

- **PSD** (positive semidefinite): $\mathbf{v}^\top A\mathbf{v} \geq 0$ for all $\mathbf{v}$
- **PD** (positive definite): $\mathbf{v}^\top A\mathbf{v} > 0$ for all $\mathbf{v} \neq \mathbf{0}$
- **NSD** (negative semidefinite): $\mathbf{v}^\top A\mathbf{v} \leq 0$ for all $\mathbf{v}$
- **ND** (negative definite): $\mathbf{v}^\top A\mathbf{v} < 0$ for all $\mathbf{v} \neq \mathbf{0}$

3.8 Examples — where PSD and NSD appear

$A^\top A$ **is PSD.** For any $\mathbf{v}$,

$$\mathbf{v}^\top A^\top A\mathbf{v} = \|A\mathbf{v}\|^2 \geq 0.$$

Covariance matrix Σ is PSD. For centered data, $\Sigma = \mathbb{E}[\mathbf{x}\mathbf{x}^\top]$, so

$$\mathbf{v}^\top \Sigma \mathbf{v} = \mathbb{E}\big[(\mathbf{v}^\top \mathbf{x})^2\big] \geq 0$$

(an average of squares).

3.8 Example — the Hessian

Reminder (Day 2). For $f : \mathbb{R}^n \to \mathbb{R}$ in C^2, the **Hessian** $H_f(\mathbf{x})$ is the matrix of **second partial derivatives**:

$$\left(H_f(\mathbf{x})\right)_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}).$$

In 2 variables: $H_f = \begin{pmatrix} f_{x_1x_1} & f_{x_1x_2} \\ f_{x_2x_1} & f_{x_2x_2} \end{pmatrix}$.

Symmetric. Mixed partials agree ($f_{x_ix_j} = f_{x_jx_i}$), so $H_f(\mathbf{x})^\mathsf{T} = H_f(\mathbf{x})$ — a symmetric matrix, so Section 3.8 applies.

Next module (Day 4). We connect H_f to **concavity**: concave $f \Leftrightarrow H_f(\mathbf{x})$ is **NSD** for all $\mathbf{x}$. Today: NSD is the sign condition on the quadratic form $\mathbf{v}^\mathsf{T} H_f(\mathbf{x})\mathbf{v}$.

3.8 Characterizations — eigenvalues and diagonal entries

Assume $A = A^{\mathsf{T}}$.

Eigenvalues (if $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$) — **necessary** conditions:

- **PSD** $\Rightarrow$ every $\lambda_i \geq 0$
- **PD** $\Rightarrow$ every $\lambda_i > 0$
- **NSD** $\Rightarrow$ every $\lambda_i \leq 0$
- **ND** $\Rightarrow$ every $\lambda_i < 0$

For symmetric A, these are also **sufficient** (equivalent to the definitions); proof omitted at Math Camp.

Diagonal entries ($A = (a_{ij})$) — **necessary** conditions: PSD $\Rightarrow a_{ii} \geq 0$; NSD $\Rightarrow a_{ii} \leq 0$. If A is **diagonal**, these sign checks are also **sufficient**.

3.8 Leading principal minors — definition

Leading principal minor. For $A \in \mathbb{R}^{n\times n}$, the k**th leading principal minor** is

$$\Delta_k = \det\big(\text{top-left } k \times k \text{ block of } A\big), \qquad k = 1, \dots, n.$$

Example (3×3)**.** $A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$:

$$\Delta_1 = a, \qquad \Delta_2 = \det\begin{pmatrix} a & b \\ d & e \end{pmatrix} = ae - bd, \qquad \Delta_3 = \det(A).$$

Only the **north-west** blocks — not e.g. the bottom-right 2×2 block unless it lines up from row 1.

3.8 Leading principal minors — PSD, PD, NSD, ND

Assume $A = A^{\mathsf{T}}$. Leading principal minors give **equivalent** tests in the strict cases; for PSD use **all** principal minors.

- **PD:** $\Delta_k > 0$ for $k = 1, \ldots, n$ (Sylvester)
- **PSD:** every **principal minor** ≥ 0 (leading alone is **not** enough)
- **NSD:** $(-1)^k \Delta_k \geq 0$ for $k = 1, \ldots, n$
- **ND:** $(-1)^k \Delta_k > 0$ for $k = 1, \ldots, n$

Proofs of principal-minor criteria omitted at Math Camp level.

3.8 2×2 checklist

$$A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \ \Delta_1 = a, \ \Delta_2 = ac - b^2.$$

- **PSD:** $a \geq 0, \ c \geq 0, \ ac - b^2 \geq 0$
- **PD:** $a > 0, \ ac - b^2 > 0$
- **NSD:** $a \leq 0, \ c \leq 0, \ ac - b^2 \geq 0$
- **ND:** $a < 0, \ ac - b^2 > 0$

Example. $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$: $a, c > 0, \ ac - b^2 = 3 > 0 \Rightarrow$ PD (matches eigenvalues $3, 1$ from Section 3.7).

3.8 Exercise: 3×3 — leading principal minors   Take home

Exercise. For

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix},$$

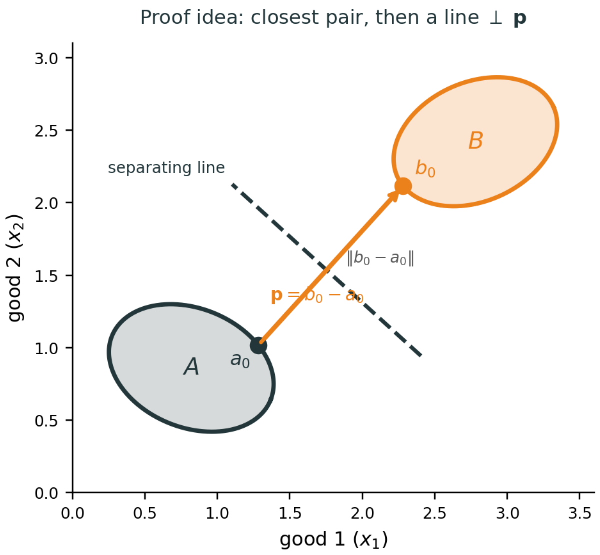
compute $\Delta_1, \Delta_2, \Delta_3$ and decide whether A is PD, PSD only, or neither.

Solution. $\Delta_1 = 2 > 0$. $\Delta_2 = \det \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = 4 - 1 = 3 > 0$.
$\Delta_3 = \det(A) = 2(4-1) + (-1)(-2) = 6 + 2 = 8 > 0$. All leading principal minors are positive $\Rightarrow$ A is **PD** (Sylvester, Section 3.8).

Appendix

Not required for Math Camp. Proofs and extra examples linked from the main slides.

Order matches main text: A.1–A.2 (Section 3.2), A.3 (3.5), A.4 (3.6), A.5–A.11 (3.7).


Proof idea: closest pair, then a line ⊥ p
separating line
B
b_0
A
a_0
$\mathbf{p} = b_0 - a_0$
$\|b_0 - a_0\|$
good 1 (x_1)
good 2 (x_2)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5

A.1 Proof. Separation when A is compact

for interest only

Claim. Disjoint nonempty convex $A, B \subset \mathbb{R}^n$ with A compact and B closed. Then $\exists\, \mathbf{p} \neq \mathbf{0}$, $\beta \in \mathbb{R}$ with $\mathbf{p} \cdot a \leq \beta \leq \mathbf{p} \cdot b$ for all $a \in A$, $b \in B$.

Step 1 — closest pair. Define $d = \inf\{\|a - b\| : a \in A,\ b \in B\} > 0$. Pick sequences $(a_n) \subset A$, $(b_n) \subset B$ with $\|a_n - b_n\| \to d$. Since A is compact, a subsequence $a_{n_k} \to a_0 \in A$. The (b_{n_k}) are bounded, so (using B closed) a further subsequence $b_{n_{k_j}} \to b_0 \in B$. Hence the infimum is **attained** at some pair $(a_0, b_0) \in A \times B$.

Not an interior point. a_0 and b_0 need only lie in A and B; they are usually on the **boundaries**, not in $\mathrm{int}(A)$ or $\mathrm{int}(B)$.

Step 2 — normal vector. Set $\mathbf{p} = b_0 - a_0 \neq \mathbf{0}$ (points from a_0 toward b_0).

A.1 Step 1 — why the subsequence argument works

for interest only

Why $a_{n_k} \to a_0 \in A$. In $\mathbb{R}^n$, **compact** $=$ closed $+$ bounded (Heine–Borel). Every sequence in a compact set has a **subsequence** converging to a limit **in that set**. So $(a_n) \subset A$ compact $\Rightarrow\ \exists\, a_{n_k} \to a_0 \in A$.

Why (b_{n_k}) is bounded. Since $\|a_{n_k} - b_{n_k}\| \to d$ and (a_{n_k}) converges, for large k,

$$\|b_{n_k}\| \leq \|b_{n_k} - a_{n_k}\| + \|a_{n_k}\| \leq d + 1 + \|a_0\| + 1.$$

(Triangle inequality: $\|b\| \leq \|b - a\| + \|a\|$.)

Why $b_{n_{k_j}} \to b_0 \in B$. A bounded sequence in $\mathbb{R}^n$ has a convergent subsequence (Bolzano–Weierstrass). Since B is **closed**, the limit b_0 lies in B.

A.1 Step 3 — why the derivative works

for interest only

Setup. Fix $a \in A$. For $t \in [0,1]$, convexity gives $a_t = a_0 + t(a - a_0) \in A$ and $g(t) = \|a_t - b_0\|^2$. Since (a_0, b_0) minimizes distance, $t = 0$ minimizes g on $[0,1]$.

Why use the derivative? A minimum at the left endpoint $t = 0$ means the slope cannot point downward: $g'(0) \geq 0$.

Differentiate. $g(t) = \|a_0 + t(a - a_0) - b_0\|^2$, so

$$g'(t) = 2\langle a_0 + t(a - a_0) - b_0,\ a - a_0\rangle, \qquad g'(0) = 2\langle a_0 - b_0,\ a - a_0\rangle \geq 0.$$

Hence $\langle b_0 - a_0,\ a - a_0\rangle \leq 0$. With $\mathbf{p} = b_0 - a_0$: $\mathbf{p} \cdot (a - a_0) \leq 0$, so $\mathbf{p} \cdot a \leq \mathbf{p} \cdot a_0$.

A.1 Proof. Separation when A is compact (continued)

Step 4 — B stays on the other side. Fix $b \in B$. For $t \in [0,1]$, set $b_t = b_0 + t(b - b_0) \in B$ and $h(t) = \|a_0 - b_t\|^2$. Since $t = 0$ minimizes h on $[0,1]$, the same derivative argument gives $\mathbf{p} \cdot b \geq \mathbf{p} \cdot b_0$ for all $b \in B$.

Step 5 — finish. Since $A \cap B = \varnothing$, the closest points satisfy $a_0 \neq b_0$. With $\mathbf{p} = b_0 - a_0$,

$$\mathbf{p} \cdot b_0 - \mathbf{p} \cdot a_0 = \mathbf{p} \cdot (b_0 - a_0) = \|b_0 - a_0\|^2 = d^2 > 0,$$

so $\mathbf{p} \cdot a_0 < \mathbf{p} \cdot b_0$. Any β with $\mathbf{p} \cdot a_0 \leq \beta \leq \mathbf{p} \cdot b_0$ works (e.g. $\beta = \mathbf{p} \cdot a_0$ gives **weak** separation: $\mathbf{p} \cdot a \leq \beta \leq \mathbf{p} \cdot b$ for all $a \in A$, $b \in B$).

A.1 Strict separation

for interest only

Weak vs. strict. From Steps 3–5: $\mathbf{p} \cdot a \leq \mathbf{p} \cdot a_0$ and $\mathbf{p} \cdot b \geq \mathbf{p} \cdot b_0$. So any β with $\mathbf{p} \cdot a_0 \leq \beta \leq \mathbf{p} \cdot b_0$ separates weakly (e.g. $\beta = \mathbf{p} \cdot a_0$).

When is separation strict? If $d = \|b_0 - a_0\| > 0$ (sets bounded apart; e.g. both compact and disjoint), then $\mathbf{p} \cdot a_0 < \mathbf{p} \cdot b_0$ with a **positive gap**. Pick β **strictly between** them, e.g. $\beta = \frac{1}{2}(\mathbf{p} \cdot a_0 + \mathbf{p} \cdot b_0)$. Then

$$\mathbf{p} \cdot a \leq \mathbf{p} \cdot a_0 < \beta < \mathbf{p} \cdot b_0 \leq \mathbf{p} \cdot b \quad \Rightarrow \quad \mathbf{p} \cdot a < \beta < \mathbf{p} \cdot b.$$

If $d = 0$ (sets touch in the limit, e.g. $A = [0, 1]$, $B = (1, 2]$), only weak separation is possible.

A.1 Proof. Remarks

for interest only

If only B is compact. Swap the roles of A and B, apply the proof, then relabel the normal vector.

Single point $B = \{b_0\}$. Let $\hat{a}$ be a point in A closest to b_0 (exists since A is closed and convex). The same calculation with $(a_0, b_0) = (\hat{a}, b_0)$ separates A from b_0 even when A is not compact.

A.2 Proof. Efficient production $\Rightarrow$ profit maximization

for interest only

Goal. $y \in Y$ technically efficient $\Rightarrow\ \exists\, \mathbf{p} \neq \mathbf{0}$ with $\mathbf{p} \cdot y \geq \mathbf{p} \cdot y'$ for all $y' \in Y$.

Assumptions (camp level). $Y \subset \mathbb{R}^n$ is **closed and convex**; Y is **bounded** (so compact); **free disposal** (if $y \in Y$ and $\tilde{y} \leq y$, then $\tilde{y} \in Y$).

Step 1 — better set. Define

$$P_y = \{y' \in \mathbb{R}^n : \ y' \geq y, \ y' \neq y\}$$

(weakly more output, strictly more in at least one good). P_y is **convex**.

Step 2 — disjointness. If $y' \in P_y \cap Y$, then y' weakly dominates y in Y, contradicting efficiency. So $P_y \cap Y = \varnothing$.

A.2 Proof. Efficient production (continued)

for interest only

Step 3 — separate. Y is compact and P_y is closed, so apply the separating hyperplane theorem (swap labels if needed so a compact set is on one side). There exist $\mathbf{p} \neq \mathbf{0}$ and β with

$$\mathbf{p} \cdot y'' \leq \beta \leq \mathbf{p} \cdot y' \quad \forall\, y'' \in Y,\ \forall\, y' \in P_y.$$

Hence $\mathbf{p} \cdot y'' \leq \mathbf{p} \cdot y'$ for all $y'' \in Y$, $y' \in P_y$.

Step 4 — profit maximization at y. For each good i, pick $y'^{(i)} = y + \varepsilon e_i \in P_y$ for small $\varepsilon > 0$ (feasible with free disposal). Then $\mathbf{p} \cdot y'' \leq \mathbf{p} \cdot y'^{(i)} = \mathbf{p} \cdot y + \varepsilon p_i$. Let $\varepsilon \downarrow 0$ to get $\mathbf{p} \cdot y'' \leq \mathbf{p} \cdot y$ for all $y'' \in Y$.

Remark. Separation gives only $\mathbf{p} \neq \mathbf{0}$. For $\mathbf{p} \geq 0$, need free disposal on all goods: if $p_i < 0$, then $y - \varepsilon e_i \in Y$ (since $\leq y$) raises profit, contradicting maximization at y.

A.3 Transpose rules — proof

for interest only

$(A^\top)^\top = A$**.** Transpose swaps rows and columns twice, so you return to A.

$(AB)^\top = B^\top A^\top$**.** Entry (i,j) of $(AB)^\top$ equals entry (j,i) of AB:

$$\left((AB)^\top\right)_{ij} = (AB)_{ji} = \sum_k a_{jk} b_{ki} = \sum_k (B^\top)_{ik}(A^\top)_{kj} = \left(B^\top A^\top\right)_{ij}.$$

Same matrix $\Rightarrow$ equal.

$(A\mathbf{x})^\top = \mathbf{x}^\top A^\top$**.** Special case with $B = \mathbf{x}$ a column $(n \times 1)$: $(A\mathbf{x})^\top = \mathbf{x}^\top A^\top$.

A.3 Inverse rules — proof

for interest only

$(AB)^{-1} = B^{-1}A^{-1}$ (both invertible). Multiply in reverse order:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = I.$$

Similarly $(B^{-1}A^{-1})(AB) = I$. So $B^{-1}A^{-1}$ is the inverse of AB.

$(A^{-1})^{-1} = A$**.** By definition of inverse, $A^{-1}A = I$, so A is the inverse of A^{-1}.

$(A^{-1})^{\mathsf{T}} = (A^{\mathsf{T}})^{-1}$**.** Transpose $A^{-1}A = I$:

$$(A^{-1}A)^{\mathsf{T}} = A^{\mathsf{T}}(A^{-1})^{\mathsf{T}} = I.$$

So $(A^{-1})^{\mathsf{T}}$ is the inverse of A^{T}.

A.4 Projection matrix — why $P_X^2 = P_X$?

for interest only

Setup. $P_X = X(X^\top X)^{-1}X^\top$ is the **projection matrix** (hat matrix); $M_X = I - P_X$ is the **annihilator matrix** (residual maker matrix).

Idempotent: $P_X^2 = P_X$ **(project once is enough).** $P_X\mathbf{y} = \widehat{\mathbf{y}}$ already lies in $\mathrm{Col}(X)$. Vectors in that yellow plane are unchanged by projecting onto it again:

$$P_X\widehat{\mathbf{y}} = \widehat{\mathbf{y}} \qquad \Longrightarrow \qquad P_X(P_X\mathbf{y}) = P_X\mathbf{y}.$$

First drop the part orthogonal to $\mathrm{Col}(X)$; a second drop removes nothing.

A.4 Dot product and transpose — why $(A\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (A^\top \mathbf{v})$?

for interest only

Goal. This is the **transpose rule** used in the symmetry proof of P_X.

Rewrite both sides as row $\times$ column. Dot product $=$ transpose the first vector, then multiply:

$$(A\mathbf{u}) \cdot \mathbf{v} = (A\mathbf{u})^\top \mathbf{v} = \mathbf{u}^\top A^\top \mathbf{v},$$

using $(A\mathbf{u})^\top = \mathbf{u}^\top A^\top$.

Other side is the same expression.

$$\mathbf{u} \cdot (A^\top \mathbf{v}) = \mathbf{u}^\top (A^\top \mathbf{v}).$$

So $(A\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (A^\top \mathbf{v})$ for every $A, \mathbf{u}, \mathbf{v}$.

Meaning: $A^\top$ moves A from the left of $\mathbf{u}$ to the right of $\mathbf{u}$ inside an inner product.

A.4 Projection matrix — why $P_X^\top = P_X$?

for interest only

Step 1 — geometric balance. $P_X\mathbf{u} \in \mathrm{Col}(X)$, so it is orthogonal to the part of $\mathbf{v}$ outside that space:

$$(P_X\mathbf{u}) \cdot \mathbf{v} = (P_X\mathbf{u}) \cdot (P_X\mathbf{v}).$$

Swapping $\mathbf{u} \leftrightarrow \mathbf{v}$ gives the same with $\mathbf{u}$ on the right. Hence

$$(P_X\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (P_X\mathbf{v}) \qquad \text{for all } \mathbf{u}, \mathbf{v} \in \mathbb{R}^n.$$

Step 2 — transpose rule. Previous slide: $(P_X\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (P_X^\top\mathbf{v})$.

Step 3 — conclude symmetry. Steps 1 and 2 give $\mathbf{u} \cdot (P_X^\top\mathbf{v}) = \mathbf{u} \cdot (P_X\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$. Fix $\mathbf{v} = \mathbf{e}_j$ and $\mathbf{u} = \mathbf{e}_i$: the jth column of $P_X^\top$ equals the jth column of P_X for every j. So $P_X^\top = P_X$.

Direct check: $(P_X)^\top = X\big((X^\top X)^{-1}\big)^\top X^\top = P_X$, using $(X^\top X)^{-1} = ((X^\top X)^\top)^{-1} = (X^\top X)^{-\top}$.

A.5 Symmetric matrices — eigenvalues are real

for interest only

Claim. If $A = A^{\mathsf{T}}$ (real symmetric), every eigenvalue λ is **real**.

Proof (setup). Let $A\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$. We allow **complex v**: over $\mathbb{C}$ eigenvalues always exist; we show $\lambda \in \mathbb{R}$.

$\mathbf{v}^{\mathsf{H}}$ = **conjugate transpose** of **v** (transpose, then conjugate each entry).

Multiply $A\mathbf{v} = \lambda\mathbf{v}$ on the left by $\mathbf{v}^{\mathsf{H}}$:

$$\mathbf{v}^{\mathsf{H}} A\mathbf{v} = \lambda\, \mathbf{v}^{\mathsf{H}}\mathbf{v}.$$

Here $\mathbf{v}^{\mathsf{H}}\mathbf{v} = \|\mathbf{v}\|^2 > 0$ (a positive real number).

A.5 Symmetric matrices — eigenvalues are real (cont.)

for interest only

Left side is real. Conjugate-transpose rule: $(\mathbf{v}^{\mathsf{H}} A\mathbf{v})^* = \mathbf{v}^{\mathsf{H}} A^{\mathsf{H}}\mathbf{v}$. A is real and symmetric, so $A^{\mathsf{H}} = A^{\mathsf{T}} = A$:

$$(\mathbf{v}^{\mathsf{H}} A\mathbf{v})^* = \mathbf{v}^{\mathsf{H}} A\mathbf{v} \quad \Longrightarrow \quad \mathbf{v}^{\mathsf{H}} A\mathbf{v} \in \mathbb{R}.$$

(A scalar equals its own conjugate $\Leftrightarrow$ it is real.)

Finish. From $\mathbf{v}^{\mathsf{H}} A\mathbf{v} = \lambda\, \mathbf{v}^{\mathsf{H}}\mathbf{v}$, divide by $\mathbf{v}^{\mathsf{H}}\mathbf{v} > 0$:

$$\lambda = \frac{\mathbf{v}^{\mathsf{H}} A\mathbf{v}}{\mathbf{v}^{\mathsf{H}}\mathbf{v}} \in \mathbb{R}.$$

Real numerator, positive real denominator $\Rightarrow \lambda \in \mathbb{R}$.

Not all symmetric eigenvalues are positive — e.g. $A = \mathrm{diag}(1, -1)$. Sign conditions (PSD/NSD) are in Section 3.8.

A.6 Equivalent linear systems

for interest only

Equivalent systems. Two linear systems (same unknowns) are **equivalent** if they have **exactly the same solutions**.

Easier equivalent system. Same solution set, but written in a form that is simpler to read — e.g. fewer redundant equations, leading variables isolated, zeros below each pivot. Row reduction replaces a system by an easier equivalent one **without changing who solves it**.

Homogeneous case. For $M\mathbf{x} = \mathbf{0}$, every equivalent system has the same **kernel** (null space)

$$\ker M = \{\mathbf{x} : M\mathbf{x} = \mathbf{0}\}.$$

Finding eigenvectors for λ is the same problem with $M = A - \lambda I$.

A.6 Elementary row operations

for interest only

Allowed operations on a matrix (or augmented matrix $[M \mid \mathbf{b}]$):

1. **Swap** two rows.
2. **Scale** one row by a **nonzero** number.
3. **Add** a multiple of one row to another row.

Each operation produces an **equivalent system** (same solution set).

Why solutions are preserved (idea). Swapping or scaling an equation does not change its truth set. If row i and row j are both satisfied, so is (row j) $+c$ (row i); conversely you can undo the combination using the other allowed ops.

RREF. Keep applying these ops until the matrix is in **reduced row echelon form**: leading 1 in each pivot row; only zero above/below each leading 1; each leading 1 lies to the right of the one above.

A.6 Pivot variables and free variables

for interest only

Once a matrix is in **RREF**, each column is either a **pivot column** or a **free column**.

Pivot column / pivot variable. A column with a **leading** 1; the matching unknown is a **pivot variable**. Its RREF row expresses the pivot variable as an **affine function of the free variables**.

Free column / free variable. A column *without* a leading 1; the matching unknown is a **free variable**. You may set each free variable to **any** value; nothing in RREF forces it.

Key step. Read each pivot row and **write each pivot variable as a function of the free variables**:

$$x_{\text{pivot}} = (\text{expression involving only free variables}).$$

Once the free variables are chosen, every pivot variable is determined; the whole solution is fixed.

A.6 Example: row reduction (non-trivial)

for interest only

Problem. Find all $\mathbf{x} \in \mathbb{R}^4$ with $M\mathbf{x} = \mathbf{0}$ for

$$M = \begin{pmatrix} 1 & 0 & 2 & 1 \\ 2 & 0 & 4 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & 2 & 0 \end{pmatrix}.$$

(Coupled equations; not block-diagonal.)

Row ops. R2 $\leftarrow$ R2$-$2R1 eliminates the duplicate x_1-row; R4 $\leftarrow$ R4$-$2R3 eliminates the duplicate x_2-row:

$$\longrightarrow \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{\text{swap R2,R3}} \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \text{(RREF)}.$$

A.6 Example: pivot variables as functions of free variables

for interest only

From the RREF above: pivot columns $1, 2 \Rightarrow$ pivot variables x_1, x_2; free columns $3, 4 \Rightarrow$ free variables x_3, x_4.

Equivalent system:

$$x_1 + 2x_3 + x_4 = 0, \qquad x_2 + x_3 = 0.$$

Solve each pivot row for its pivot variable:

$$x_1 = -2x_3 - x_4, \qquad x_2 = -x_3, \qquad x_3, x_4 \in \mathbb{R} \text{ free.}$$

Every solution is fixed once (x_3, x_4) is chosen.

A.6 Basis of the kernel from free variables

for interest only

General solution.

$$\mathbf{x} = x_3 \begin{pmatrix} -2 \\ -1 \\ 1 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

So $\ker M = \operatorname{span}\{\mathbf{b}_1, \mathbf{b}_2\}$ with $\dim \ker M = 2$ (two free variables).

Recipe (basis of $\ker M$**).** For each free variable in turn:

1. set that free variable $= 1$;
2. set all other free variables $= 0$;
3. compute pivot variables from the RREF rows.

Here: $x_3 = 1$, $x_4 = 0$ gives $\mathbf{b}_1$; $x_3 = 0$, $x_4 = 1$ gives $\mathbf{b}_2$ (the two columns above).

Check: $M\mathbf{b}_1 = \mathbf{0}$, $M\mathbf{b}_2 = \mathbf{0}$.

A.7 When eigenvalues are not real

for interest only

The characteristic equation $\det(A - \lambda I) = 0$ may have **no real roots**. Then A has **no real eigenvectors** (no real direction kept by scaling alone).

Example: 90° **rotation** (from earlier slides), $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$:

$$\det(A - \lambda I) = \lambda^2 + 1 = 0 \quad \Rightarrow \quad \lambda = \pm i \text{ (not real)}.$$

Every nonzero vector is rotated off its own line, so no $\mathbf{v} \in \mathbb{R}^2 \setminus \{\mathbf{0}\}$ satisfies $A\mathbf{v} = \lambda\mathbf{v}$.

Over $\mathbb{C}$. Allow complex λ and $\mathbf{v}$; the same recipe (characteristic polynomial, then $(A - \lambda I)\mathbf{v} = \mathbf{0}$) works. For **real** A, non-real eigenvalues come in **conjugate pairs** $\lambda, \bar{\lambda}$.

Existence over $\mathbb{C}$: appendix (**for interest only**). When a real eigenbasis fails, see A.9–A.11 for QR and SVD (**for interest only**).

A.8 Orthogonal and orthonormal matrices

for interest only

Orthonormal columns. A square matrix $Q \in \mathbb{R}^{n\times n}$ is **orthogonal** if

$$Q^{\top} Q = I \qquad (\text{equivalently } QQ^{\top} = I).$$

Column i is $\mathbf{q}_i$; then $\mathbf{q}_i^{\top}\mathbf{q}_j = \delta_{ij}$ (orthonormal).

Geometric meaning. Q is a **rigid motion** of $\mathbb{R}^n$ (rotation and/or reflection):

$$\|Q\mathbf{x}\| = \|\mathbf{x}\|, \qquad (Q\mathbf{x})^{\top}(Q\mathbf{y}) = \mathbf{x}^{\top}\mathbf{y}.$$

Lengths and angles are preserved; $Q^{-1} = Q^{\top}$.

Terminology. “Orthonormal matrix” usually means the same as orthogonal (unit-length orthonormal columns).

A.8 QR and SVD (brief)

for interest only

Same spirit as eigen-decomposition: rewrite A so the action is easy to read.

- **QR** ($A = QR$): Q has orthonormal columns, R is upper triangular. Used for stable least squares and picking independent columns (appendix).
- **SVD** ($A = U\Sigma V^{\top}$): always exists; U, V orthogonal, Σ diagonal (singular values). Scales along orthogonal directions — works even when real eigenvectors do not exist (e.g. rotation).

More detail on main slide "Decompositions — QR, eigen, SVD" (for interest only). Existence: appendix, appendix (**for interest only**).

A.9 Existence of eigenvalues and eigenvectors (over $\mathbb{C}$)

for interest only

Claim. Every $A \in \mathbb{C}^{n \times n}$ has at least one eigenpair $(\lambda, \mathbf{v})$ with $\mathbf{v} \neq \mathbf{0}$.

Step 1 (characteristic polynomial). $p_A(\lambda) = \det(A - \lambda I)$ is a polynomial in λ of degree n.

Step 2 (eigenvalue exists). By the **fundamental theorem of algebra**, $p_A(\lambda) = 0$ has n roots in $\mathbb{C}$ (counting multiplicity). Pick one root λ.

Step 3 (eigenvector exists). $p_A(\lambda) = 0$ means $\det(A - \lambda I) = 0$, so $A - \lambda I$ is **singular**. Hence $(A - \lambda I)\mathbf{v} = \mathbf{0}$ has a nontrivial solution $\mathbf{v} \neq \mathbf{0}$.

Real A: non-real roots come in conjugate pairs $\lambda, \bar{\lambda}$.

A.10 Existence of QR — what the decomposition means

for interest only

Claim. Every $A \in \mathbb{R}^{m\times n}$ can be written as $A = QR$: Q has **orthonormal columns**, R is **upper triangular**.

Setup. Write $A = [\mathbf{a}_1\, \mathbf{a}_2\, \mathbf{a}_3\, \cdots]$ — a **series of column vectors** 1, 2, 3, . . . They may overlap (not perpendicular, different lengths).

Goal (Gram–Schmidt). Build new **clean directions** $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3, \ldots$ (perpendicular, unit length) so that

vector j is built only from $\mathbf{q}_1, \ldots, \mathbf{q}_j$.

The numbers r_{ij} in R record **how much** of each vector goes along each direction.

A.10 Existence of QR — vectors 1 and 2

for interest only

Vector 1. Take the first column, normalize it — that is the first clean direction:

$$\mathbf{q}_1 = \frac{\mathbf{a}_1}{\|\mathbf{a}_1\|}.$$

Vector 1 lies entirely along $\mathbf{q}_1$.

Vector 2. Project $\mathbf{a}_2$ onto $\mathbf{q}_1$, subtract, and keep the **residual** as the new direction:

$$\mathbf{a}_2 = \underbrace{(\mathbf{q}_1^\top \mathbf{a}_2)\,\mathbf{q}_1}_{\text{part along } \mathbf{q}_1} + \underbrace{\left(\mathbf{a}_2 - (\mathbf{q}_1^\top \mathbf{a}_2)\mathbf{q}_1\right)}_{\perp\ \mathbf{q}_1}.$$

Normalize the residual to get $\mathbf{q}_2$. **Key:** vector 2 uses **only** $\mathbf{q}_1$ and $\mathbf{q}_2$: $\mathbf{a}_2 = r_{12}\mathbf{q}_1 + r_{22}\mathbf{q}_2$.

A.10 Existence of QR — vectors 3, 4, . . .

for interest only

Vector 3. Project $\mathbf{a}_3$ onto $\mathbf{q}_1$ **and** $\mathbf{q}_2$, subtract both overlaps. Normalize what's left $\Rightarrow \mathbf{q}_3$. Vector 3 uses **only** $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$.

Vector 4, . . . Same pattern: for vector k, subtract its projections onto all previous clean directions, normalize the residual $\Rightarrow \mathbf{q}_k$.

General step.

$$\mathbf{a}_k = \sum_{j=1}^{k-1} (\mathbf{q}_j^\top \mathbf{a}_k)\, \mathbf{q}_j + (\text{residual} \perp \text{all previous } \mathbf{q}_j).$$

So $\mathbf{a}_k = r_{1k}\mathbf{q}_1 + \cdots + r_{kk}\mathbf{q}_k$ — no $\mathbf{q}_\ell$ with $\ell > k$ needed.

A.10 Existence of QR — reading R and $A = QR$

for interest only

What is r_{ij}? From $\mathbf{a}_j = \sum_i \mathbf{q}_i r_{ij}$: entry r_{ij} is **how much of vector j lies along direction i**:

$$r_{ij} = \mathbf{q}_i^\top \mathbf{a}_j.$$

Why upper triangular? Vector j is built from $\mathbf{q}_1, \ldots, \mathbf{q}_j$ only, so $r_{ij} = 0$ for $i > j$. Diagonal r_{jj} = length of the **residual** (the genuinely new part).

Existence. Run the process on all columns; stack $\mathbf{q}_j$ into Q and r_{ij} into R. Then $A = QR$.

A.11 Existence of SVD — intuition

for interest only

Dimensions. $A \in \mathbb{R}^{m \times n}$ maps $\mathbb{R}^n \to \mathbb{R}^m$: $\mathbf{x} \in \mathbb{R}^n \Rightarrow A\mathbf{x} \in \mathbb{R}^m$.

Why SVD? Eigenvectors need $A\mathbf{v} = \lambda\mathbf{v}$ (same direction in and out) — fine for square A with a full eigenbasis. Fails for rotation (no real eigenvectors) and when $m \neq n$ (input and output live in different spaces).

Idea. Find **unit** input directions $\mathbf{v}_i$ ($\|\mathbf{v}_i\| = 1$) and unit output directions $\mathbf{u}_i$ ($\|\mathbf{u}_i\| = 1$) with

$$A\mathbf{v}_i = \sigma_i \mathbf{u}_i, \qquad \sigma_i \geq 0.$$

Input direction $\mathbf{v}_i$ maps to output direction $\mathbf{u}_i$, stretched by σ_i (no mixing).

Picture. Unit sphere in $\mathbb{R}^n \to$ ellipsoid in $\mathbb{R}^m$. $\mathbf{v}_i =$ input axes; $\mathbf{u}_i =$ where they land; $\sigma_i =$ axis lengths. Factorization: $A = U\Sigma V^{\mathsf{T}} =$ rotate input (V), scale (Σ), rotate output (U).

A.11 Existence of SVD — derivation (Step 1)

for interest only

Step 1 — input directions ($\mathbb{R}^n$). Diagonalize $A^\top A$ ($n \times n$): orthonormal eigenvectors $\mathbf{v}_i$ with $\|\mathbf{v}_i\| = 1$, eigenvalues $\lambda_i \geq 0$. Set $\sigma_i = \sqrt{\lambda_i} = \|A\mathbf{v}_i\|$ (stretch along unit input $\mathbf{v}_i$).

Remark. λ_i is real (A.5, appendix) and $\lambda_i \geq 0$ (Section 3.8) because $A^\top A$ is symmetric PSD. So $\sigma_i = \sqrt{\lambda_i}$ is real and ≥ 0; it is **not** an eigenvalue of A.

A.11 Existence of SVD — derivation (Steps 2–3)

for interest only

Step 2 — output directions ($\mathbb{R}^m$). Goal: $A\mathbf{v}_i = \sigma_i \mathbf{u}_i$ with $\|\mathbf{u}_i\| = 1$. From Step 1: $\mathbf{v}_i$ is **normalized** ($\|\mathbf{v}_i\| = 1$) and $\sigma_i = \|A\mathbf{v}_i\|$.

Apply A to the unit input $\mathbf{v}_i$: the vector $A\mathbf{v}_i \in \mathbb{R}^m$ points the right way but has length σ_i. Normalize it:

$$\mathbf{u}_i = \frac{1}{\sigma_i} A\mathbf{v}_i \quad \Longrightarrow \quad A\mathbf{v}_i = \sigma_i \mathbf{u}_i, \quad \|\mathbf{u}_i\| = 1.$$

Step 3 — assemble. $V = [\mathbf{v}_1 \ \cdots]$, $U = [\mathbf{u}_1 \ \cdots]$, $\Sigma = \mathrm{diag}(\sigma_1, \ldots) \Rightarrow A = U\Sigma V^\top$.